

INORGANIC CHEMISTRY

TIME : 30 Min.

SECTION-I (i) : (Maximum Marks : 30)

- This section contains **TEN** questions. **Marking scheme** : +3 (-1)
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.

1. A divalent monoatomic cation (atomic number Z) is isoelectronic of CO_2 and has (Z + 2) neutron. What is ionic mass of divalent cation.

- (A) 48 (B) 50 (C) 44 (D) none

2. x^- , y^{2-} and z^{3-} are isotonic and isoelectronic. Thus increasing order of atomic number of x, y and z is

- (A) $x < y < z$ (B) $z < y < x$ (C) $x = y = z$ (D) $z < x < y$

3. Which of the following sets of quantum numbers represents the highest energy of an electron?

- (A) $n = 4, l = 0, m = 0, s = +1/2$ (B) $n = 4, l = 2, m = 1, s = +1/2$
(C) $n = 5, l = 0, m = 1, s = -1/2$ (D) $n = 4, l = 1, m = 0, s = -1/2$

4. A certain negative ion A^{2-} has 18 neutrons and 18 electrons. What is the mass number of most abundant isotope of A

- (A) 34 (B) 36 (C) 32 (D) none

5. An element 'X' with mass number 81 contains 31.7% more neutrons as compared to protons, then what is the representation of element 'X'.

- (A) ${}_{38}^{81}\text{X}$ (B) ${}_{35}^{81}\text{X}$ (C) ${}_{34}^{81}\text{X}$ (D) none

6. Among the following pair of orbitals which orbits will experience the larger effective nuclear charge

- (i) 2s and 3s (ii) 4d and 4f (iii) 3d and 3p
(A) 2s, 4d and 3p (B) 3s, 4f and 3p (C) 2s, 4f and 3d (D) 2s, 4d and 3d

7. Which of the following statements are **CORRECT**?

- (I) A large number of orbitals are possible in an atom.
(II) Qualitatively these orbitals can be distinguished by their size, shape and orientation
(III) An orbital of smaller size means there is more probability of finding the electron near the nucleus
(IV) Each orbital is designated by four quantum numbers

- (A) Only IV (B) I and II (C) I, III and IV (D) I, II and III

8. Which of the following statements are **INCORRECT**?

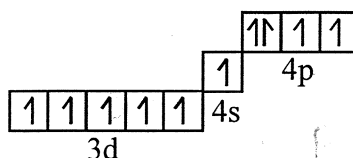
- (I) Orbit and orbital are synonymous
(II) There are n^2 orbitals in shell, all the orbitals of a given value of n constitute a single shell of an atom
(III) Hydrogen and hydrogen like systems (eg. He^+ , Li^{2+}), principal quantum number alone determines the energy and size of the orbitals
(IV) The principal quantum number, n, is a positive integer with values of 1, 2, 3, etc.

- (A) Only I (B) I and II (C) I and IV (D) I, III and IV

9. Given this set of quantum number for a multi electronic atom $2, 0, 0, \frac{1}{2}$ and $2, 0, 0, -\frac{1}{2}$. What is the next higher energetic allowed set of n and l quantum numbers

- (A) $n = 2, l = 0$ (B) $n = 2, l = 1$ (C) $n = 3, l = 0$ (D) $n = 3, l = 1$

10. In the following electronic configuration, some rules have been violated.



- (I) Hund's rule (II) Pauli's exclusion principle (III) Aufbau principle
(A) I and II (B) I and III (C) II and III (D) I, II and III

Section-I (ii) : (Maximum Marks : 20)

- This section contains **FIVE** questions. **Marking scheme** : +4 (0)
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.

11. Which of the following statements are **CORRECT** ?

- (A) In a p-orbital the number of nodal planes equal to the azimuthal quantum number
(B) The electronic energy of the H-atom depends only on n and is independent of l and m
(C) Those substances in which all electrons spin are paired slightly repelled by an external magnetic field and are said paramagnetic
(D) Those substances in which one or more electrons have unpaired spins are relatively strongly attracted to magnetic field and are said to be diamagnetic

12. Which of the following statements are **CORRECT** ?

- (A) The possible values of all the four quantum numbers for the 19th electron of Cr is $n = 4$, $l = 0$, $m = 0$ and $s = +1/2$
(B) The possible values of all the four quantum numbers for the 21st electron of Sc is $n = 3$, $l = 2$, $m = -2$ and $s = +1/2$
(C) The possible values of all the four quantum numbers for the 7th electron of N is $n = 2$, $l = -1$, $m = 1$, $s = -1/2$
(D) The possible values of all the four quantum numbers for the electron of H is $n = 0$, $l = 0$, $m = 0$, $s = -1/2$

13. Which of the following paramagnetic ion have same number of unpaired electrons.

- (A) Fe^{+3} (B) Mn^{+2} (C) Co^{+4} (D) Cr^{+}

14. Which of the following statement is **CORRECT** for d_{z^2} orbital.

- (A) It has value of $l = 2$
(B) It has four lobes
(C) It has two nodal cone
(D) It is axial orbital

15. Which of the following configuration is incorrect according to Hund's rule

- (A) $\uparrow\downarrow\uparrow\uparrow\uparrow$ (B) $\uparrow\downarrow\uparrow\uparrow$ (C) $\uparrow\downarrow\uparrow\uparrow\uparrow\uparrow$ (D) $\uparrow\downarrow\uparrow\downarrow$

SECTION-IV : (Maximum Marks : 20)

- This section contains **FIVE** questions. **Marking scheme** : +4(-1)
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive

1. Find the number of orbitals, which have atleast one common nodal plane with d_{xz} .

2. Find the difference in number of electrons of s-orbitals in Fe and Ni^{+2} .

3. Find the value of x , if Mn^{+x} has magnetic moment value is 3.87 B.M.

4. Number of electrons Arrangement

1	$\uparrow$	—	—
2	$\uparrow$	$\uparrow$	—
3	$\uparrow$	$\uparrow$	$\uparrow$
6	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$

Find the value of $x + y + z + w$.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

5. Find the value of Z_{eff} of last electron of Ni excluding decimal value.

INORGANIC CHEMISTRY

TIME : 30 Min

SECTION-I (i) : (Maximum Marks : 18)

- This section contains **SIX** questions. **Marking scheme** : +3 (–1)
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.

- The number of d-electrons in Cr^{+3} is not equal to the number of valence electrons in which of the following :-
 (A) p-electrons in 'P' (B) d-electrons in 'V'
 (C) d-electrons in Mn^{+2} (D) p-electrons in As
- Among the following pair of elements which are liquid at room temperature :-
 (A) Ga, Ge (B) Hg, Br (C) Te, Cl (D) Hg, Ge
- Which of the following sets of quantum number is **NOT** possible ?

	n	l	m	s
(A)	3	2	-2	$+\frac{1}{2}$
(B)	2	0	0	$-\frac{1}{2}$
(C)	3	3	-1	$+\frac{1}{2}$
(D)	4	3	-3	$-\frac{1}{2}$
- The Z_{eff} for :
 2s electron in F
 3d electron in Zn
 3s electron in Cr^{+3} are respectively :-
 (A) 3.8 , 21.15 , 11.25 (B) 5.2 , 21.15 , 12.75
 (C) 5.2 , 8.85 , 12.75 (D) 4.6 , 12.75 , 11.25
- Select the **CORRECT** order of atomic radius ?
 (A) $\text{B} < \text{Al} > \text{Ga}$ (B) $\text{Si} > \text{Ge} > \text{Sn}$ (C) $\text{I} < \text{S} \approx \text{C}$ (D) $\text{Li} < \text{Na} < \text{Be}$
- Which of the following statement is **INCORRECT** for d_{z^2} orbital.
 (A) It has value of $l = 2$ (B) It has four lobes
 (C) It has two nodal cone (D) It is axial orbital

Section-I (ii) : (Maximum Marks : 24)

- This section contains **SIX** questions. **Marking scheme** : +4 (0)
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.

- Which of the following configuration is **CORRECT** according to Hund's rule

(A) $\uparrow\downarrow \uparrow \uparrow \square \square$ (B) $\uparrow \downarrow \uparrow \square \square$ (C) $\uparrow\downarrow \uparrow \uparrow \uparrow \uparrow$ (D) $\downarrow \downarrow \square \downarrow \square$

- Which two electrons of fluorine atom can have same azimuthal quantum number :-

(A) 1st , 2nd (B) 1st , 3rd (C) 3rd , 5th (D) 5th , 6th

9. Which two elements have same n, l, m_l, m_s value for last e^- ?
(A) Ni and Pd (B) Zn and Mn (C) Cu and Ag (D) Cr and Cu
10. Which of the following is/are heterogeneous mixture :-
(A) Mixture of iron and sulphur (B) Milk
(C) Mixture of alcohol and water (D) Air
11. An atom has 10 electrons, 10 neutrons and 10 protons. Which of the following is/are NOT isotonic with this atom ?
(A) $^{40}\text{Ca}_{20}$ (B) $^{36}\text{Ar}_{18}$ (C) $^{20}\text{Ne}_{10}$ (D) $^{12}\text{C}_6$
12. In which of the following arrangements, the **CORRECT** order according to the property indicated against it is ?
(A) $\text{IE}_1 : \text{O} > \text{N} > \text{C} > \text{B}$ (B) Vander Waal's radius : $\text{Cl} > \text{F} > \text{Br} > \text{I}$
(C) Metallic radius : $\text{Rb} > \text{K} > \text{Na} > \text{Li}$ (D) Ionic size : $\text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

SECTION-I(iii) : (Maximum Marks : 18)

- This section contains **THREE** paragraphs.
- Based on each paragraph, there will be **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
0 In all other cases

PARAGRAPH FOR Q. NO. (13 TO 14)

Three elements L, M, N [having $20 < \text{atomic number} \leq 30$] have following properties :-

- (i) 'L' have 5 electrons with $l = 2$
(ii) 'M' have total 15 electrons with $n = 3$
(iii) 'N' have spin only magnetic moment value 'zero' in its +2 oxidation state
13. Element which will have highest atomic number amongst these will be :-
(A) L (B) M
(C) N (D) Information is incomplete
14. Element which will have highest spin only magnetic moment in +2 oxidation state will be :-
(A) L (B) M
(C) N (D) All three have identical value

PARAGRAPH FOR Q. NO. (15 TO 16)

Element	I.E_1 (KJ/mol)	I.E_2 (KJ/mol)
Li	520	7296
Mg	737	1450
K	419	3069

15. Out of these three elements, maximum I.E_3 value will be of :-
(A) Mg (B) K (C) Li (D) Either Mg or K
16. Element Ca ($Z = 20$) can have I.E_1 value as :-
(A) 398 KJ/mol (B) 590 KJ/mol (C) 820 KJ/mol (D) 920 KJ/mol

PARAGRAPH FOR Q. NO. (17 TO 18)

Four element X, Y, Z and M have atomic number respectively 10, 11, 17 and 18

17. Which of the above element will have highest $I.E_1$ amongst these :-
(A) X (B) Y (C) Z (D) M
18. Element amongst these which have highest metallic character
(A) Y (B) Z (C) X (D) M

SECTION-II : Matrix-Match Type

This Section contains **2 question**. Each question has **four statements** (A, B, C and D) given in **Column I** and **four statements** (P, Q, R & S) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. Match the following :-

Column-I (Element)

- (A) Na
(B) Cl
(C) F
(D) He

Column-II (Property)

- (P) Highest $I.E_1$ amongst these
(Q) Lowest $I.E_1$ amongst these
(R) $I.E_1$ is more than only one element amongst these
(S) Smallest size among the halogens
(T) Belongs to IVth period

2. Matrix match :-

Column - I (Element)

- (A) Cr
(B) Zn
(C) Ne
(D) S

Column-II (Property)

- (P) Fullfilled 'd' subshell is present
(Q) Half filled 'd' subshell is present
(R) Fullfilled valence shell
(S) Two more electrons are required to achieve inert gas configuration
(T) Six more electrons are required to achieve d^{10} configuration

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INORGANIC CHEMISTRY

TIME : 30 Min.

SECTION-I (i) : (Maximum Marks : 9)

- This section contains **THREE** questions. **Marking scheme** : +3 (–1)
 - Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- Which of the following having maximum number of unpaired electron -
 (A) Na^+ (B) N^{3-} (C) Fe^{3+} (D) Cr^{3+}
 - An ion Cr^{+n} has the magnetic moment equal to 5.92 BM. The value of n is
 (A) 5 (B) 1 (C) 2 (D) none
 - What is the lowest value of n that allows 'g' orbitals to exist ?
 (A) 4 (B) 5 (C) 6 (D) 7

Section-I (ii) : (Maximum Marks : 20)

- This section contains **FIVE** questions. **Marking scheme** : +4 (0)
 - Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- An element have electronic configuration as $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^7$.
 Correct statement regarding element is -
 (A) Element must be Co
 (B) Element belongs to group no. '9' according to the long form of periodic table
 (C) Maximum no. of electrons in element having $m = +1$ are 6
 (D) Element have magnetic moment $\sqrt{15}$ B.M.
 - Select the correct energy order for following orbitals:
 (A) $4f < 3d$ (B) $3p < 4s$ (C) $3d > 4s$ (D) $4p > 5s$
 - Among following which pairs have same magnetic moment
 (A) $\text{Cr}^+, \text{Mn}^{2+}$ (B) $\text{Cr}^+, \text{Fe}^{2+}$ (C) $\text{Fe}^{3+}, \text{Mn}^{2+}$ (D) $\text{Co}^{+2}, \text{Cr}^{+3}$
 - Select the incorrect statement (s) from the following options :-
 (A) The nodal plane of p_x is identical with that of p_y orbital
 (B) $d_{x^2-y^2}$ orbital has two nodal plane which are xz and yz planes
 (C) p_x, p_y and p_z orbitals have the same value of ' ℓ '
 (D) $d_{xy}, d_{x^2-y^2}$ and d_{z^2} orbitals have the different orientation in space but have the same value of ' m '
 - Select the correct statement (s) -
 (A) Value of σ for 3s electron of 'Cl' is 9.15
 (B) Number of unpaired electrons in Fe^{+3} is 5
 (C) Cr^{+3} will have same number of unpaired electrons as present in ground state configuration of 'V'
 (D) Number of electrons having $\ell = 2$ in Mn^{+2} is 3

SECTION-I(iii) : (Maximum Marks : 9)

- This section contains **ONE** paragraphs. **Marking scheme** : +3 (0)
- Based on each paragraph, there will be **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct.

Paragraph for Q.9 to Q.11

The general electronic configuration of outer most and penultimate shell is given as $(n-1)s^2 (n-1)p^6 (n-1)d^x ns^2$. Then for an element with $n = 4$ and $x = 7$.

- The number of protons present in the divalent cation of the element of above configuration is :-
 (A) 25 (B) 26 (C) 27 (D) 28
- The element is isoelectronic with which of the following species :-
 (A) Fe^- (B) Ni^+ (C) Cu^{2+} (D) All of these

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11. The number of unpaired electron in the trivalent cation of the given element in isolated gaseous state is :-
 (A) 0 (B) 3 (C) 4 (D) 1

SECTION-II : Matrix-Match Type

This Section contains **ONE** question. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. Match the column -

Column-I

- (A) l
 (B) s
 (C) m
 (D) n

Column-II

- (P) Indicates the distance from nucleus
 (Q) It can have only two possible values.
 (R) Possible values are 0 to $(n - 1)$
 (S) Indicates the possible orientations in space.
 (T) Possible values are $+l$ to $-l$ including 'Zero'

SECTION-IV : (Maximum Marks : 40)

- This section contains **TEN** questions. **Marking scheme** : +4(-1)
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive

1. A node is a point or region in space where the wave function has zero amplitude (point at which the electron density is zero). In general, the number of radial nodes = $n - l - 1$. Find the radial node for 1s orbital.
 2. Find the number of orbitals which have electronic density along the axis

$d_{xy}, d_{yz}, d_{zx}, d_{x^2-y^2}, d_{z^2}, p_x, p_y, p_z$

3. Find the $(n + l)$ value of 6g orbital.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

4. How many elements would be in the second period of the periodic table if the spin quantum number m_s could have the value $-1/2, 0, +1/2$.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

5. Find the number of atoms which have equal Z_{eff}
 Li, Na, K, Rb

6. Find the number of Homogeneous mixtures :

Air, Hydrogen in palladium, Ethyl alcohol in water, Mercury in amalgamated zinc,
 Sugar in water, Alloy, Water + oil, Chalk in water.

7. Find the number of unpaired electron present in Xe^+ in $\text{Xe}^+[\text{PtF}_6]^-$

8. Find total no. of orbitals in nickel which have $|m| \leq 1$ and at least one electron is present, where 'm' is magnetic quantum number.

(Given your ans. as sum digits for example. If your ans is 57 then $5 + 7 = 12$ and $1 + 2 = 3$,

9. Three isolated gaseous ions are M^{+1} , M^{2+} and M^{3+} , which are having the magnetic moment value of 5.92 BM, 4.89 BM and 2.82 BM respectively. Calculate the total number of unpaired electron in the above three species.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

10. Find the atomic number of the element (M) present in 3d-series for which the M^{2+} gives the magnetic moment of 5.92 B.M.

[If the answer is 17, then represent as $1 + 7 = 8$ is the answer is 08.]

SECTION-I (i) : (Maximum Marks : 18)

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- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases

Handwritten notes:
S
b > c
Ar
Ar > c > r

1. Select **CORRECT** order of size :

- (A) ~~Al~~ > C > F (B) C > Al > F (C) Al > F > C (D) C > Al > F

2. Select which of the following have atomic number between 111 to 115.

- (A) Uns (B) Uus
(C) ~~Unq~~ (D) Uuh

Handwritten note: $1s^2 2s^2 2p^4$

3. In which reaction, electronic configuration is changed from $1s^2 2s^2 2p^2$ to $1s^2 2s^2 2p^3$

- (A) $C_{(g)} + e^- \rightarrow N_{(g)}$ (B) $C_{(g)} + e^- \rightarrow C_{(g)}^-$
(C) $N_{(g)} \rightarrow N_{(g)}^+ + e^-$ (D) $Si_{(g)} + e^- \rightarrow Si_{(g)}^-$

4. In which series, number of protons and size both are increasing on moving from first to fourth :

- (A) $O^{2-}, F^-, Na^+, Mg^{2+}$ (B) ~~$O^{2-}, S^{2-}, Se^{2-}, Te^{2-}$~~
(C) $Ti^{+4}, Ti^{3+}, Ti^{2+}, Ti^+$ (D) I, Br, Cl, F^-

5. When shielding is considered according to Slater rules, the last electron of oxygen is held with about% of the nuclear charge as expected from +8 nucleus charge on last electron.

Select **CORRECT** options to fill in the black space.

- (A) ~~57%~~ (B) 20% (C) 90% (D) 82%

6. Consider following sets of quantum number

$$n = 4 ; l = \dots ; m = -1 ; s = +1/2$$

Missing digit is required to characterise :-

- (A) Shape of orbital of 'M' shell
(B) Orientation of orbitals in p-subshell of 4th shell
(C) Size of orbitals as compare to 6p orbital
(D) ~~Shape of the directional orbitals of N shell~~

(ii) : (Maximum Marks : 16)

- This section contains **FOUR** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

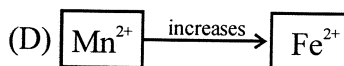
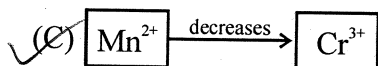
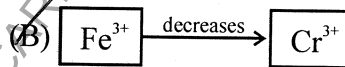
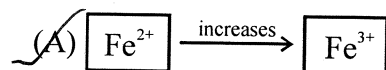
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 In all other cases

7. Atom(s) in periodic table in which any subshell has half filled configuration

- ☒ (A) Cr (Z = 24) ☒ (B) Mn (Z = 25) ☒ (C) P (Z = 15) (D) Si (Z = 14)

8. Select **CORRECT** for paramagnetism



9. Select **CORRECT** option(s) for the long form of periodic table.

☒ (A) Elements having similar outer electronic configuration of outer shell are arranged in vertical groups

☒ (B) 3rd & 4th period have 8 and 18 elements respectively

☒ (C) 14 elements of sixth period are placed in f-block

☒ (D) Period number corresponds to the highest principal quantum number (n) in electronic configuration of elements

10. Upto 3rd shell

☒ (A) $l = 0$ is observed for 3 different orbitals ✓

(B) $l = 1$ is observed for 2 different orbitals

☒ (C) $l = 2$ is observed for 5 different orbitals ✓

☒ (D) $l = 0$ is observed for 3 different subshells

1s
2s 2p
3s 3p 3d

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **Two questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

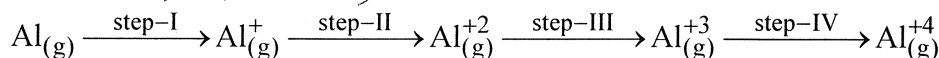
Paragraph-1

Consider ionisation energy of all elements of 3rd period in the periodic table and give the answer of following questions :-

11. Which of the following have largest difference in magnitude -
 (A) IE_1 of Cl and IE_1 of Ar (B) IE_1 of Na and IE_1 of P
 (C) IE_1 of Na and IE_1 of Cl (D) IE_1 of Ar and IE_1 of Na
12. Which element have highest IE_2 in 3rd period.
 (A) Phosphorus (B) Argon (C) Sodium (D) Magnesium

B
Ar
 $3s^2 3p^1$

Paragraph-2

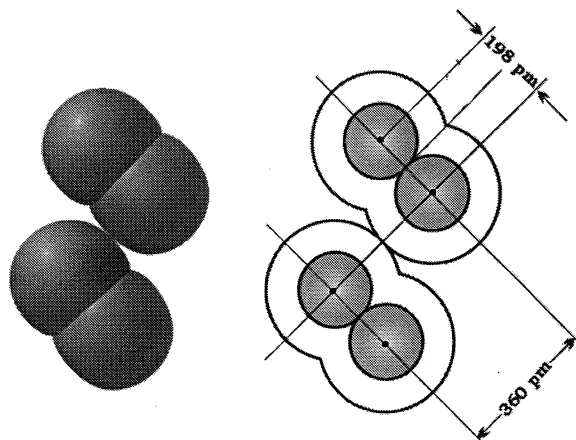


Consider the following change in a atom of aluminium to form tetrapositive ion of Al and give the following answer.

13. Sum of enthalpy step-II and step-III represent
 (A) $IE_3 + IE_4$ of Al (B) $IE_1 + IE_2 + IE_3$ of Al
 (C) $IE_3 + IE_2 - IE_1$ of Al (D) $IE_3 + IE_2$ of Al
14. Which set of quantum number is NOT observed for any electron after first ionization of Al.
 (A) $n = 3, l = 0, m = 0, s = +1/2$ (B) $n = 3, l = 1, m = 0, s = +1/2$
 (C) $n = 2, l = 1, m = +1, s = -1/2$ (D) $n = 2, l = 1, m = +1, s = -1/2$

Paragraph-3

Consider two closest Cl_2 molecules to find covalent and Vander Waals radii of Cl atom in chlorine molecule . The inner circles in figure correspond to the size of the chlorine atom .



$\frac{180}{99} = 1.81$

15. Select **CORRECT** data which represent covalent radius of chlorine :
 (A) 180 pm (B) 198 pm (C) 360 pm (D) 99 pm
16. What is the difference of vander wall radius and covalent radius of chlorine atom :
 (A) 81 pm (B) 162 pm (C) 18 pm (D) none of the above

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SECTION-IV : (Maximum Marks : 16)

- This section contains **FOUR** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :

+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases

1. Number of common nodal planes in $3d_{xy}$ and $4d_{xy}$. (2)
2. What is the value of n for given change in an atom A.
 $A_{(g)}^{n+} \longrightarrow A_{(g)}^{(n+1)+} + e^{-} \quad \Delta H = + IE_6$ (5)
3. Consider Slater rules and find number of electrons of N (nitrogen atom) which is/are NOT shielded by 2p electrons. (2)
4. An M^{7+} ion have total four electron(s) with $n = 3, l = 2, s = +1/2$ in its ground state and No electron in $n = 3, l = 2, m = 0$ orbital. Find unpaired electron(s) in M^{+2} ion. (4)

1 1 1 1

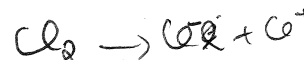
INORGANIC CHEMISTRY

TIME : 30 Min

SECTION-I (i) : (Maximum Marks : 6)

- This section contains **TWO** questions. **Marking scheme : +3(-1)**
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.

1. Which enthalpy is **not** associated with following change
Molecular chlorine $\longrightarrow$ chloride ion



- (A) Bond dissociation enthalpy ✓
(B) First ionisation enthalpy
(C) First electron gain enthalpy
(D) None

2. Shielding constant of last electron of which element will be maximum -

- (A) H (B) H⁻ (C) He (D) Li

SECTION-I(ii) : (Maximum Marks : 20)

- This section contains **FIVE** questions. **Marking scheme : +4(0)**
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.

3. Which of the following ion having radius greater than F⁻

- (A) H⁻ (B) Cl⁻ (C) Br⁻ (D) I⁻

4. Which of the following subshell(s) of outer shell is fully filled in Hg (Z = 80)

- (A) s (B) p (C) d (D) f

5. Which of the following set consists of only isoelectronic species :

- (A) N³⁻, O²⁻, Ne, Na⁺ (B) NO₃⁻, SiO₄⁴⁻, CO₃²⁻ (C) Hg²⁺, Pb⁴⁺ (D) H, He⁺, Li²⁺, Be³⁺

6. Which of the following processes involves absorption of energy ?

- (A) S_(g) + e⁻ → S_(g)⁻ (B) Xe_(g) + e⁻ → Xe_(g)⁻ (C) O_(g)⁻ + e⁻ → O_(g)²⁻ (D) Cl_(g)⁻ → Cl_(g) + e⁻

7. In which of the following arrangements the order which is **CORRECT** according to the property indicated against it.

- (A) Al³⁺ < Mg²⁺ < Na⁺ < F⁻ : Increasing ionic size ✓
(B) B < C < N < O : Increasing first ionisation enthalpy —
(C) I < Br < F < Cl : Increasing electron gain enthalpy (magnitude only) ✓
(D) Li < Na < K < Rb : Increasing metallic radius

1s
2s
3s 3p 3d
4s 4p 4d
5s 5p 5d
6s

1s² 2s² 2p⁶ 3s² 3p⁶
4s² 3d¹⁰ 4p⁶ 5s²
4d¹⁰ 5p⁶ 6s²
4f¹⁴ 5d¹⁰

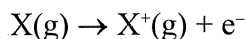
B C N O
Al Si P S
Zn Ga Ge As Se
Cd In Sn Sb Te
Hg Pb Bi Po

SECTION-I(iii) : (Maximum Marks : 12)

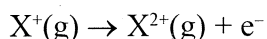
- This section contains **TWO** paragraphs. **Marking scheme** : +3(0)
- Based on each paragraph, there will be **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct.

Passage for Q.no. 8 and 9

A quantitative measure of the tendency of an element to lose electron is given by its **Ionization Enthalpy**. It represents the energy required to remove an electron from an isolated gaseous atom (X) in its ground state. In other words, the first ionization enthalpy for an element X is the enthalpy change ($\Delta_i H$) for the reaction depicted.



The ionization enthalpy is expressed in units of kJ mol^{-1} . We can define the second ionization enthalpy as the energy required to remove the second most loosely bound electron; it is the energy required to carry out the reaction.



8. Largest energy is required for which change ?



9. Consider following statements for the two uncharged gaseous species

Specie - 1 = $1s^2, 2s^1$ **Specie - 2** = $1s^2, 2p^1$

- (I) specie-1 is ground state, specie-2 is excited state of same atom ✓
(II) specie-1 is excited state, specie-2 is ground state of same atom
(III) ionization of specie-1 is easier as compared to specie-2
(IV) ionization of specie-2 is easier as compared to specie-1 ✓

Select **CORRECT** statement(s)

- (A) I and III (B) II and III ~~(C) I and IV~~ (D) II and IV

Passage for Q.no. 10 and 11

Wave function for electrons in atoms called atomic orbitals. Atomic orbitals are labelled as three quantum numbers n, l, m .

10. Which orbitals have two directional lobes, which can point in three possible directions?

- (A) All orbitals which have $n = 2$ (B) All orbitals which have $l = 2$
(C) All orbitals which have $m = -1/0/+1$ ~~(D) All orbitals which have $l = 1$~~

11. Select invalid set of quantum number for any electron in ground state electronic configuration of potassium

- (A) $n = 4, l = 0, m = 0, s = +1/2$ (B) $n = 3, l = 1, m = 0, s = -1/2$
~~(C) $n = 3, l = 2, m = 0, s = +1/2$~~ (D) $n = 2, l = 1, m = -1, s = -1/2$

SECTION-I(iv) : (Maximum Marks : 9)

- This section contains **THREE** questions. **Marking scheme** : +3(−1)
- Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

12. List-I (Element)

(P) N

(Q) O

(R) Be

(S) B

List-II (ΔH_{IE_1} in kJ / mol)

(1) +800

(2) +900

(3) +1300

(4) +1400

Code :

	P	Q	R	S
(A)	2	3	1	4
(B)	1	2	3	4
(C)	4	3	2	1
(D)	1	4	2	3

13. List-I

(P) Sn ($Z = 50$)

(Q) V ($Z = 23$)

(R) Rb ($Z = 37$)

(S) Y ($Z = 39$)

List-II

(Convention of 1–18 group in long periodic table)

(1) 5th group

(2) 4th period

(3) 1st group

(4) 5th period

Code :

	P	Q	R	S
(A)	4	1, 2	3, 4	4
(B)	3, 4	1, 2	1, 4	4
(C)	4	2, 3	4	3, 4
(D)	1, 4	3, 4	1	2, 4

14. List-I

(Elements having atomic number)

(P) (11)

(Q) (19)

(R) (37)

(S) (55)

List-II

(1) IE is maximum

(2) $Z_{\text{eff}} = 2.2$ (on last electron)

(3) Size largest

(4) Position is next to Kr

Code :

	P	Q	R	S
(A)	1	1, 2	1, 2	3, 4 ×
(B)	1, 2	2, 4	2, 4	2, 3
(C)	1, 2	2	2, 4	2, 3
(D)	2, 3	2	2, 4	2, 3 ×

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SECTION-IV : (Maximum Marks : 40)

- This section contains **TEN** questions. **Marking scheme : +4(-1)**
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive

1. Select pair which have higher electronegativity difference as compared to difference of Cs & F. (0)
Na & I; Li & F; H & F; I & Cl; Ca & O; Ba & S; C & Si
2. Among the given elements, number of species which are isodiaphers of each other (4)
 ${}_{92}\text{U}^{238}$, ${}_{88}\text{Ra}^{230}$, ${}_{89}\text{Ac}^{232}$, ${}_{90}\text{Th}^{234}$
3. Calculate the difference in magnitude of Z_{eff} on 4s and 3d-electron of Sc. (6)
4. Total number of orbitals that are to be filled completely before entering into the 6s subshell for filling of electrons (Add the digits till you get a single digit integer) (9)
5. If spin quantum number can have three possible values, then the maximum number of elements in 6th period is 'a'. Give the value of $\left(\frac{a}{6}\right)$. (8)
6. Calculate the maximum number of electrons in nitrogen atom for which $n + s = 1.5$ (5)
7. The spin magnetic moment of Cr^{+3} is $\sqrt{x}$ B.M. The value of $\left(\frac{x}{3}\right)$ is : (9)
8. Among the following species, how many have their ionic size greater than S^{2-} . (2)
 O^{2-} , P^{3-} , Cl^{-1} , S^{-1} , Ca^{+2} , K^{+}
9. From the given elements find how many elements have lower value of I.E. from Cs. (2)
Li, Na, K, Rb, Ba
10. Find the number of elements which does not have isotope. (1)
H, Cl, O, C, N

SECTION-I (i) : (Maximum Marks : 48)

- This section contains **SIXTEEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases

H
4
2

- Total number of subshells possible for $n + l = 7$ are-
(A) 3 (B) 5 (C) 4 (D) 2
- Choose the correct option among the following statements (T = True, F = False)-
(i) d and f subshell have very high shielding effect as compared to s and p subshell **F**
(ii) among 3rd period elements IE_2 of Na is maximum **T**
(iii) Electron gain enthalpy of inert gases is greater than zero i.e. an endothermic process **T**
(iv) Al is a typical as well as bridge element
(v) Maximum degeneracy of a hypothetical ion Na^{+10} in 2nd excited state is 3. **F**
(A) FTTFT (B) TFFTF **X** (C) FTTTF (D) FTTTTT
- Which of the following represent correct order for atomic or ionic radius
(i) $Co \approx Ni < Cu < Zn$ **✓** (ii) $Sc < Y \approx La$ **X**
(iii) $Ti < Zr \approx Hf$ **✓** (iv) $V < Nb > Ta$
(v) $Mg < Na < Ca < K$
(A) (i), (ii), (iii) **✓** (B) (i), (iii), (iv) (C) (i), (ii), (iv) **X** (D) (i), (iii), (v) **✓**
- Two elements have set of quantum numbers of their last filled subshell of valence shell as follows :
'X' $\rightarrow n = 4$ $\ell = 1$ **4 p** 'Y' $\rightarrow n = 5$ $\ell = 2$ **5 d**
Then which statement is **INCORRECT**
(A) Element 'Y' has greater atomic number than atomic number of element 'X' **✓**
(B) Element 'X' has 10 electrons in 'd' subshell **✓**
(C) Element 'X' is not a transition element **✓**
(D) Atomic size of element 'Y' is equal to the atomic size of element 'X' **✓**
- Suppose an orbital may accommodate three electrons, then estimate the number of elements in 2nd period
(A) 3 (B) 6 **8** (D) 12
- According to Moseley, a straight line graph is obtained on plotting :
(A) ν Vs Z (B) ν^2 Vs Z (C) $\sqrt{\nu}$ Vs Z (D) $\frac{1}{\nu}$ Vs Z
Where, ν is frequency and Z is atomic number

7. Select the **INCORRECT** match of species against its property

Species	Property
(A) Fe	Spin multiplicity = 5
☒ (B) Cr	Spin only magnetic moment = 6.93 BM
(C) Cu	Diamagnetic
(D) Co	Spin only magnetic moment = 3.87 BM

(5)(7) $\sqrt{63}$

3x3
4x

(3)

8. Lanthanide contraction means

- (A) Contraction of atom of lanthanum element due to poor shielding d-subshell electron
(B) Contraction of atom of lanthanum element due to high shielding of d-subshell electron
☒ (C) Contraction of atom of element after lanthanum due to poor shielding of f-subshell electron
(D) Contraction of atom of element before lanthanum due to poor shielding of f-subshell electron

9. Two elements A and B have values of electronegativity respectively are 4 and 1.2 then the compound A-B will be

- (A) Predominantly covalent (B) Predominantly ionic
(C) % covalent character less than 50% ☒ (D) Both (B) and (C)

10. The correct relation given by Pauling to calculate electronegativity of an element X is (with respect to hydrogen). Given EN of 'X' > EN of 'H'

(A) $EN(X) - EN(H) = 0.208 \sqrt{BE_{H-X} - \sqrt{BE_{X-X} \times BE_{H-H}}}$ where bond energies (B.E.) are in kJ / mole

(B) $EN(X) - EN(H) = 0.102 \sqrt{BE_{H-X} - \sqrt{BE_{X-X} \times BE_{H-H}}}$ where bond energies (BE) are in kcal / mole

☒ (C) $EN(X) - EN(H) = 0.102 \sqrt{BE_{H-X} - \sqrt{BE_{X-X} \times BE_{H-H}}}$ where bond energies (BE) are in kJ / mole

(D) Both (A) and (C) are correct

11. Electronic configuration of the Vanadium in the compound where it produces spin magnetic moment of 1.73 B.M. is

- (A) $[Ar] 4s^2 3d^3$ (B) $[Ar] 4s^2 3d^0$ ☒ (C) $[Ar] 4s^0 3d^1$ (D) $[Ar] 4s^0 3d^3$

12. For electronic configuration $ns^2 (n-2)f^{1-14} (n-1)d^{0-1}$, if $n = 7$, then the element belongs to

- (A) Lanthanide (B) Actinide
(C) Inner transition element ☒ (D) Both (B) and (C)

13. From the given information of element M



$IE_1 (M) = y \text{ kJ / mole}$

$IE_1 (M^{+2}) = z \text{ kJ / mole}$

The value of $IE_2 (M)$ will be

- (A) $(x + y + z) \text{ kJ / mole}$ (B) $(x - y - z) \text{ kJ / mole}$
(C) $(x - y + z) \text{ kJ / mole}$ (D) $z - (x + y) \text{ kJ / mole}$

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14. For the hydroxide of two elements A and B i.e AOH and BOH, if ΔEN (A-O) bond is lesser than ΔEN of (B-O) bond. Then which is incorrect statement regarding this (ΔEN = difference between electronegativity of element and oxygen) ?
- (A) BOH reacts more readily with an acid than AOH ✓
(B) AOH reacts more readily with an acid than BOH ✓
(C) BOH has more basic strength than AOH ✓
(D) Both (A) and (C) are correct statements
15. Values of Formal charge (F.C) on three oxygen atoms O(1), O(2) and O(3) respectively in O_3 molecule will be
- $O \leftarrow O = O$
- (1) (2) (3)
- (A) 0, 0, 0 (B) +1, -1, 0 (C) -1, +1, 0 (D) -1, 0, +1
16. If Z-axis is the internuclear axis, then overlapping among which orbitals will form σ bond ?
- (A) $d_{xy} + d_{xy}$ (B) $d_{xz} + d_{z^2}$ (C) $d_{xz} + p_z$ (D) $d_{z^2} + p_z$

(ii) : (Maximum Marks : 20)

- This section contains **FIVE** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 In all other cases

17. Which of the following options is/are incorrect according to the given properties -

- (A) $C < O < N < F \rightarrow 1^{st} \text{ I.P.}$ ✓
(B) $F < Cl < Br < I \rightarrow \text{Electronegativity}$ ✓
(C) $S < O < F < Cl \rightarrow \text{Electron affinity order}$ ✓
(D) $C \approx S \approx I \rightarrow \text{Electronegativity}$ ✓

G
S
C

18. Correct statement among the following is/are

- (i) IE_1 of Tl is less than that of In due to its larger atomic size ✓
(ii) $4d_{yz}$ and $4p_z$ have at least one common nodal plane ✓
(iii) Diagonal relationship is shown by all elements of 2nd and 3rd period. ✓
(iv) Cl has highest electron affinity in periodic stable ✓
(v) $I.P_1$ of Ga > $I.P_1$ of Al ✓

- (A) (ii), (iv), (v) (B) (i), (iv), (v)
(C) (iii), (iv), (v) (D) All are correct

19. Out of the following which represent group of amphoteric oxides -

- (A) Na_2O , SnO , Al_2O_3 (B) SnO , SnO_2 ✓
(C) PbO , PbO_2 , ZnO (D) BeO , Ga_2O_3 , Al_2O_3 ✓

20. The correct options represent correct relationship between various scales of electronegativity

(A) $E.N. (Pauling) = \frac{E.N. (Mulliken)}{2.8}$

(B) $E.N. (Allred Rochow) = E.N. (Pauling) - 0.744$

(C) $\frac{E.N. (Mulliken)}{2.8} - 0.744 = E.N. (Allred Rochow)$

(D) $E.N. (Pauling) = E.N. (Allred Rochow) - 0.744$

21. Which of the following are exception to Lewis octet rule ?

(A) Hypovalent molecules

(B) Hypervalent molecule

(C) Odd electron species

(D) None of these

(iii) Paragraph Type

This section contains **TWO paragraphs** each describing theory, experiment, data etc. **Two questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph For Q. No. 22 to 23

The size of any species depends on various factors such as nature of charge, magnitude of charge / oxidation state, effective nuclear charge, electronic configuration etc.

Based on the above information, answer the following questions.

22. Select the order of size which are correct?

(A) $O > O^- > O^{2-}$

(B) $B > Be > Li$

(C) $Mg > Mg^+ > Mg^{2+}$

(D) $Sc < Zn$

23. It has been observed that size of Indium (In) and Thallium (Tl) have very small difference this can be attributed to :-

(A) Actinoid contraction

(B) Lanthanoid contraction

(C) Both (A) and (B)

(D) Scandide Contraction

Paragraph For Q. No. 24 to 25

Ionisation energies of unknown elements are given below :

Element	IE (in kcal / mol)		
	I	II	III
M	209	548	919
N	100	735	1101
O	119	1092	1653
P	1500	2017	2320

24. Which pair of elements represent same group ?

(A) M, N

(B) N, O

(C) O, P

(D) M, P

25. Which amongst them is expected to be a noble gas ?

(A) M

(B) N

(C) O

(D) P

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SECTION-II : Matrix-Match Type

This Section contains **TWO** questions. Each question has **four** statements (A, B, C and D) given in **Column I** and **five** statements (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the following column :

Column-I

(Molecules)

- (A) ICl P K
(B) ClCl Q R
(C) NH₃ P R
(D) N₂ T Q S X R

Column-II

(Characteristic)

- (P) Polar bond(s)
(Q) Non polar bond(s)
(R) Sigma bond(s)
(S) Pie bond(s)
(T) Number of lone pair in molecule > one

2. Match the following column :

Column-I

(Atom)

- (A) Se S Q
(B) Br S
(C) Xe T
(D) B P R

Column-II

(Characteristic)

- (P) have vacant p-orbital in outer shell
(Q) Can have covalency = two
(R) Can have covalency = three
(S) 4th shell is outer most shell
(T) Produce MF₄ type fluorides molecule (M = given atom)

SECTION-IV : (Maximum Marks : 20)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :

+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases

1. Out of the following the number of correct statements are -

- (i) AOH > BOH > COH [Order of basic nature if EN values of A, B and C are 1.5, 2.8 and 3.2 respectively] F
(ii) N₂O > NO > N₂O₃ > N₂O₄ > N₂O₅ [Order of acidic strength] F
(iii) Difference between values of σ (shielding constant) of Be²⁺ and B³⁺ is 0.35 3
(iv) H₂O > H₂S > H₂Se > H₂Te (Order of basic strength) F
(v) O²⁻ > F⁻ > Na⁺ > Mg²⁺ > Al³⁺ (Order of $\frac{ne}{z}$, n = number of electrons) 4
(vi) B > Tl > Ga > Al (Order of I.P.) X

2. Value of electronegativity of oxygen using Allred-Powell scale. Covalent radius of oxygen is 1 Å.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

3. If values of all quantum numbers remains as such instead of spin quantum number which have 4 values. Then the sum of all the numerical answers in the given statement is, till we get single digit. (n = principal quantum number)

- (i) Total number of orbitals in a subshell (only for f sub-shell)
(ii) Total number of sub-shells in an orbit (only for n = 2)
(iii) Total number of electrons in an orbital (only of 5d)
(iv) Total number of electrons in an orbit (only in 3rd orbit)
(v) Total number of orbitals in an orbit (only for n = 4)

4. Out of the following representation regarding IE and EGE/EA are correct for an element M.

where ;

IE = Ionisation energy

EGE = Electron gain enthalpy

EA = Electron affinity

- (i) IE_1 of M(g) = EA_1 of M(g)
(ii) IE_1 of M⁺(g) = $|EA_1$ of M⁺(g)|
(iii) IE_1 of M⁺(g) = $|EGE_1$ of M²⁺(g)|
(iv) IE_1 of M³⁺(g) = $|EA_1$ of M⁴⁺(g)|
(v) Energy used to convert $M \rightarrow M^{3+} + 3e^- = (IE_1 + IE_2 + IE_3)$ of M
(vi) Total enthalpy used in process $M^- \rightarrow M^{2+} + 3e^-$ is $(IE_1 + IE_2)$ of M

5. Which out of the following options are incorrect ?

According to the mentioned properties.

- (i) $LiCl < NaCl < KCl < RbCl$ (% covalent character)
(ii) Maximum covalency of halogen including F which can be achieved is 7.
(iii) IP_1 of ion $M^{+2} > EA_1$ of M^{+3} . (IP = ionisation potential, EA = electron affinity)
(iv) $S > Se > Te > O$ (order of EA)
(v) $Li < Be < B < C$ (order of electronegativity)
(vi) $Mg^{+2} < Na < F^-$ (order of ionic size)
(vii) $Li^+ > Na^+ > K^+$ (order of hydrated size)
(viii) $NaCl > MgCl_2 > AlCl_3$ (order of lattice energy)

SECTION-I (i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. There is **NO** oxygen with formal charge -1 in :
(A) O_3 (B) HNO_3 (C) HNO_2 (D) None 1/4 m
2. If Br - Br bond length is 228 pm. Select which data is related to bond length of HBr :-
(A) 74 pm (B) 141 pm (C) 267 pm (D) 92 pm
3. Which pair of atomic numbers represent s-block elements ?
(1) 3, 12 (2) 6, 12 (3) 9, 17 (4) 7, 15
4. Select the impossible set of quantum numbers

n	l	m	s
(1) 4	0	-1	$+\frac{1}{2}$
(2) 3	2	+2	$-\frac{1}{2}$
(3) 3	1	0	$-\frac{1}{2}$
(4) 2	1	-1	$+\frac{1}{2}$

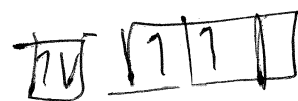
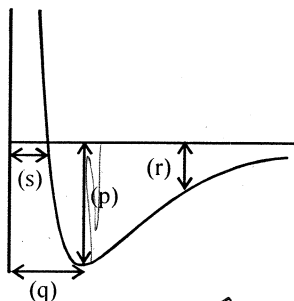


5. Select the order of size which are correct?
(1) $O > O^- > O^{2-}$ (2) $B > Be > Li$
(3) $Mg > Mg^+ > Mg^{2+}$ (4) $Sc < Zn$

(ii) : (Maximum Marks : 16)

- This section contains **FOUR** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 In all other cases

6. Given $\text{H}_{2(g)} + 435.8 \text{ kJ/mole} \rightarrow \text{H}_{(g)} + \text{H}_{(g)}$
Which is/are **CORRECT** for potential energy curve against internuclear distance for the formation of H_2



(A) $p = 435.8 \text{ kJ/mole}$

(C) $r = \text{B.E. of } \text{H}_2$

(B) $q = \text{Covalent radius of } \text{H}_2$

(D) $s = \text{B.L. of } \text{H}_2$

7. $\text{PF}_5 + \text{F}^- \rightarrow \text{Product}$

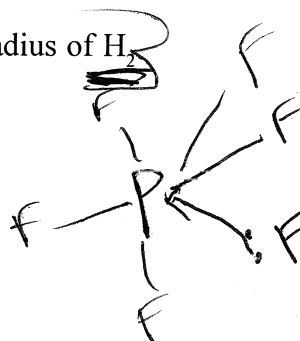
Select **CORRECT** about given reaction.

(A) F^- is lewis base and PF_5 is lewis acid

(B) Vacant orbital of phosphorus is used to form co-ordinate bond

(C) Product have total 6 sigma bonds

(D) One lone pair on phosphorus in product



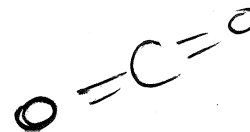
8. Select **CORRECT** about CO and CO_2 both

(A) Formal charge on carbon is +1 and zero respectively

(B) Number of L.P. on carbon is one and zero respectively

(C) Oxidation number of carbon is +2 and +4 respectively

(D) Number of π bonds by carbon is same in both



9. An atom which produce 4 covalent bonds in excited state, atom can be.

(A) C

(B) Si

(C) S

(D) Cl

(iii) Paragraph Type

This section contains **TWO paragraphs** each describing theory, experiment, data etc. **Two questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph For Q. No. 10 to 11

The effective nuclear charge (Z_{eff}) takes into account the interelectronic repulsions in multielectron atoms. Slater's rule help us as in evaluating Z_{eff} for various species. Apply Slater's rules to answer the following questions

10. Z_{eff} for 3d electron of Zn = x

Z_{eff} for 3d electron of Zn^{2+} = y

Z_{eff} for 4s electron Zn = z

Choose the **CORRECT** option(s)

(A) $x > z$

(B) $y > z$

(C) $x = y$

(D) All of these

11. Z_{eff} for valence electron of Ga = 'a'
 Z_{eff} for valence electron of Al = 'b'
 Z_{eff} for valence electron of Ca = 'c'
 Choose the **CORRECT** option(s)

(A) $a > c$

(B) $a > b$

(C) $b < c$

☒ Both (A) and (B)

Paragraph For Q. No. 12 to 13

Electron affinity and electronegativity are very closely related, yet they are very different

12. Identify the only **INCORRECT** statement

(A) Be has almost zero electron affinity ✓

(B) Between F and Cl, formation of Cl^- is more exothermic than formation of F^- ✓

(C) IE_1 of Cl^- = electron affinity of Cl ✓

☒ (D) Electronegativity depends only on electron affinity

13. Which of the following processes is exothermic ?

☒ (A) $\text{O} \rightarrow \text{O}^-$

(B) $\text{O}^- \rightarrow \text{O}^{2-}$

(C) $\text{S}^- \rightarrow \text{S}^{2-}$

(D) $\text{Mg}^+ \rightarrow \text{Mg}^{+2}$

SECTION-II : Matrix-Match Type

This Section contains **TWO** question. Each question has **four** statements (A, B, C and D) given in **Column I** and **five** statements (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. **Column-I**

(Molecules)

(A) H_2

(B) O_2

(C) O_3

(D) AlCl_3

Column-II

(Characteristic)

(P) Each atom have octet

(Q) Each atom has duplete of electron

(R) Vacant orbital in outer most shell of any atom

(S) Zero formal charge on all atoms

(T) Have π -bond

2. **Column-I**

()

(A) σ -bond

(B) π -bond

(C) δ -bond

(D) No bond formation

Column-II

(X-axis is overlapping axis)

(P) $s-p_x$ overlapping

(Q) p_y-p_y overlapping

(R) p_y-d_{xy} overlapping

(S) $d_{yz}-d_{yz}$ overlapping

(T) p_x-p_z overlapping

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SECTION-IV : (Maximum Marks : 16)

- This section contains **FOUR** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

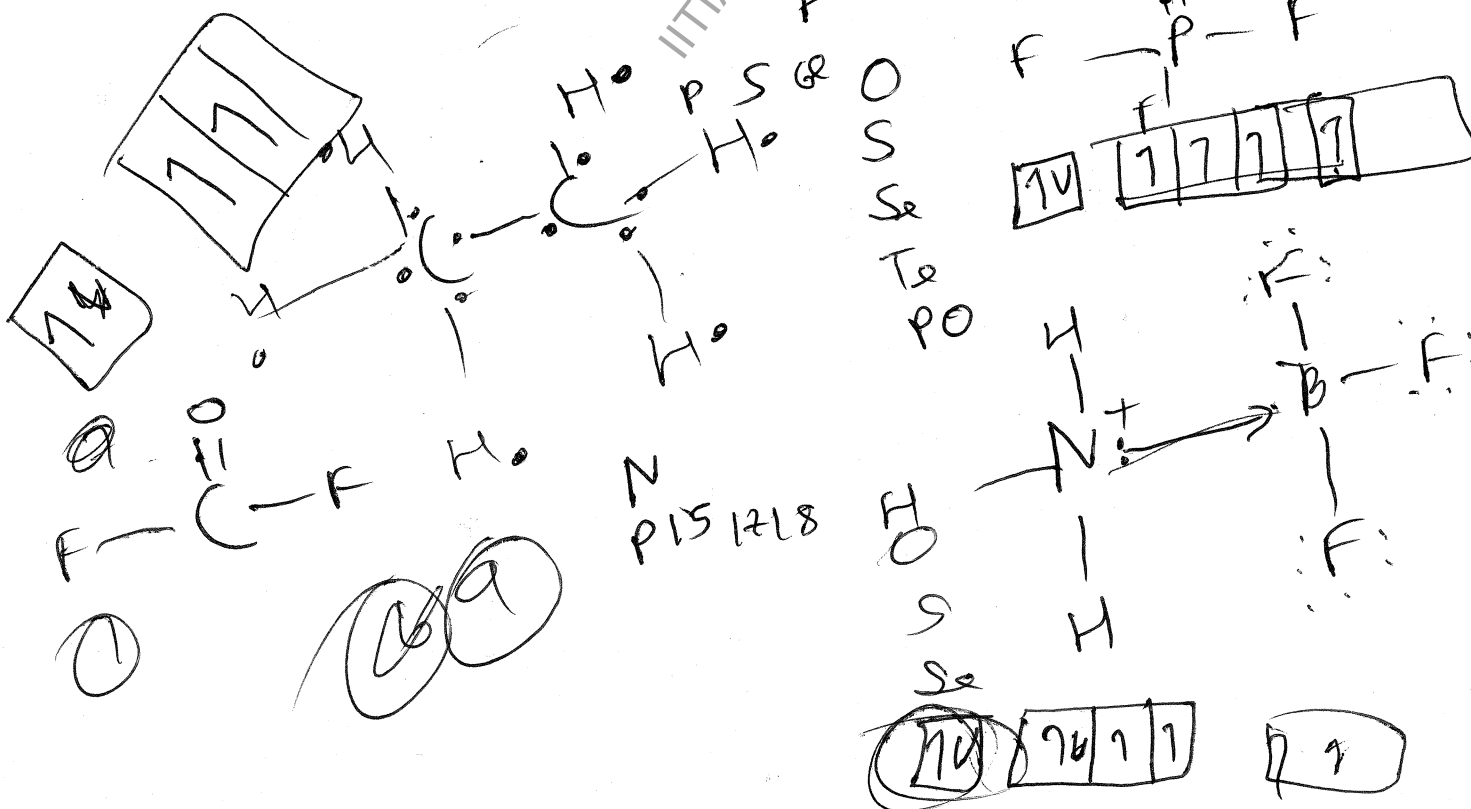
- Find number of bond pair formed by Se when selenium have only one lone pair on it. (4)
- Total number of orbitals of valence shell are involve in bonding in C_2H_6 . (5)

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

- Find number of lone pair in the product of given reaction $NH_3 + BF_3 \rightarrow$ Product. (9)
- What is the atomic number of element which have least negative electron gain enthalpy out of P, S, Cl, F. (15)

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

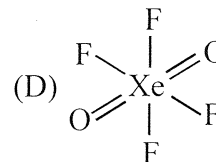
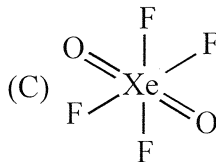
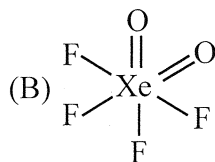
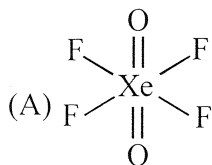
- Find sum of the π -bond(s) in COF_2 and POF_3 . (2)



SECTION-I (i) : (Maximum Marks : 36)

- This section contains **12** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

- Which of the following molecule/ion is planar and polar both :-
(A) $\text{NO}_3^\ominus$ (B) $\text{NO}_2^\ominus$ (C) $\text{NO}_2^\oplus$ (D) NH_3
- If dipole moment of $\text{NX}_3 > \text{NY}_3$ and bond angle of $\text{NX}_3 > \text{NY}_3$ then which of the following is correct (N = nitrogen)
(A) 'X' may be 'F' and 'Y' may be hydrogen
(B) 'X' may be 'H' and 'Y' may be 'F'
(C) 'X' may be 'F' and 'Y' may be 'Cl'
(D) 'X' may be 'Cl' and 'Y' may be 'F'
- Choose the element which show maximum number of covalencies out of the given elements.
(A) F (B) N (C) C (D) Cl
- Out of C_2H_6 , C_2H_4 and C_2H_2 . Compound which have highest C-C bond length is :-
(A) C_2H_4 (B) C_2H_2
(C) C_2H_6 (D) All have equal C-C bond length
- If internuclear axis is z-axis then which combination will form π -bond :-
(A) $s + p_y$ (B) $p_y + d_{xz}$ (C) $p_y + d_{yz}$ (D) $p_y + p_z$
- According to VSEPR theory which of the following structure of XeO_2F_4 is **INCORRECT**



- Which of the following order is correct against the property indicated :-
(A) $\text{PH}_4^\oplus < \text{PCl}_4^\oplus < \text{PBr}_4^\oplus$ (bond angle)
(B) $\text{BF}_3 < \text{NF}_3 < \text{NH}_3$ (dipole moment)
(C) $\text{CCl}_4 < \text{B(OH)}_3 < \text{PCl}_5$ (number of valence electrons used for bonding by central atom)
(D) $\text{CH}_4 < \text{CCl}_4 < \text{CBr}_4$ (order of bond polarity)
- Which of the following have different shape from the others :-
(A) NOF_3 (B) XeO_4 (C) SOCl_2 (D) $\text{BF}_4^\ominus$

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9. Select the correct statement about the reaction
- $$\text{BF}_3 + \text{NH}_3 \rightarrow \text{BF}_3 \cdot \text{NH}_3$$
- (A) Octet of 'N' is incomplete in product
 (B) Octet of boron is complete in product
 (C) During the reaction total number of sigma bonds remain same in the reactant as well as in the product
 (D) Type of π -bond between boron and nitrogen is $2p_\pi - 2p_\pi$
10. Total number of lone pair-bond pair repulsions at 120° in XeO_2F_2 is (Neglecting distortion) :-
 (A) 3 (B) 5 (C) 2 (D) 1
11. Which of the following order of bond angle is **CORRECT**.
 (A) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$ (B) $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$
 (C) $\text{OF}_2 < \text{H}_2\text{O} < \text{Cl}_2\text{O}$ (D) $\text{SiF}_4 < \text{SiCl}_4 < \text{SiBr}_4 < \text{SiI}_4$
12. Molecule in which central atom has sp^3d^2 hybridization is present.
 (A) IF_7 (B) IO_6^{-5} (C) XeF_2 (D) XeO_4

(ii) : (Maximum Marks : 20)

- This section contains 5 questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
 +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
 0 In all other cases

13. Which of following statement is/are **CORRECT** for ClO_2F_3 ?
 (A) In hybridisation central atom uses its d_{xy} orbital
 (B) In π -bond formation central atom uses its 'p' orbital
 (C) The shape of molecule is see-saw
 (D) The molecule is non-planar
14. Which of the following order is/are **CORRECT** with respect to the given property ?
 (A) $\text{XeF}_4 > \text{XeF}_2$: dipole moment order
 (B) $\text{CH}_3^+ > \text{CH}_3^-$: Bond angle
 (C) $2p - 2p > 2s - 2p$: σ bond strength
 (D) $\text{NF}_3 > \text{NH}_3$: lewis basic character
15. Which of the following species have same shape :-
 (A) BrF_5 (B) XeF_5^+ (C) TeCl_4 (D) ICl_4^-
16. Which of the following species are planar as well as polar.
 (A) I_3^+ (B) NO_2^- (C) PCl_3 (D) BO_3^{3-}
17. Bond angles which are associated with sp^3d^3 hybridization.
 (A) 90° (B) 120° (C) 180° (D) 72°

(iii) Paragraph Type

This section contains **ONE** paragraphs each describing theory, experiment, data etc. **Two** questions relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Passage (For Q.No.18 & 19)

Successive ionization energy of an element increases gradually due to increasing in Z_{eff} . By analyzing data of successive ionization energy one can predict the number of valence electrons.

18. The successive ionization energy data (in kJ/mole) of element is given. Predict the group number in periodic table.

IE_1	IE_2	IE_3	IE_4	IE_5	IE_6	IE_7	IE_8
789,	1580,	3230,	4360,	16100,	19800,	23800	29300
(A) 14 th		(B) 15 th		(C) 16 th		(D) 17 th	

19. First and third ionization energy of Na are 496 kJ/mole and 6910 kJ/mole respectively. The second ionization energy of Na is :

(A) 9540 kJ/mole (B) 300 kJ/mole (C) 621 kJ/mole (D) 4560 kJ/mole

(iv) : (Maximum Marks : 6)

- This section contains **2** questions. **Marking scheme** : +3(−1)
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

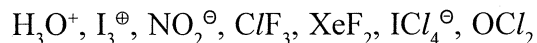
20. **List I** (Compounds)
(P) CO
(Q) XeO₃
(R) C/F₃
(S) SO₂Cl₂
- List II** (**CORRECT** characteristics)
(1) Pyramidal geometry
(2) Maximum number of atoms present in one plane is 4
(3) σ -bond and π -bond as well as co-ordinate bond.
(4) Imperfect tetrahedral geometry
- | | | | |
|-------|-----|-----|-----|
| (P) | (Q) | (R) | (S) |
| (A) 2 | 3 | 4 | 1 |
| (B) 3 | 1 | 2 | 4 |
| (C) 1 | 2 | 3 | 4 |
| (D) 4 | 3 | 2 | 1 |
21. **List I** (Species)
(P) SO₃
(Q) [SiF₆]^{2−}
(R) XeO₃F₂
(S) BF₃
- List II** (**CORRECT** characteristics)
(1) Hypovalent
(2) Axial and equatorial bonds are present
(3) $p_{\pi}-p_{\pi}$ and $p_{\pi}-d_{\pi}$ bonds are present
(4) Hybridisation of central atom is sp^3d^2
- | | | | |
|-------|-----|-----|-----|
| (P) | (Q) | (R) | (S) |
| (A) 2 | 3 | 4 | 1 |
| (B) 2 | 4 | 3 | 1 |
| (C) 3 | 2 | 4 | 1 |
| (D) 3 | 4 | 2 | 1 |

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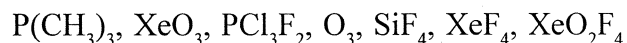
SECTION-IV : (Maximum Marks : 20)

- This section contains **5** questions. **Marking scheme : +4(−1)**
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive

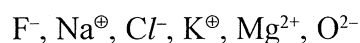
1. Among the following total number of planar molecules/ions is.



2. Find the total number of following molecules which are polar.



3. How many ions among the following have higher value of their ionic radius as compared to H^- .



4. Determine the maximum number of electrons in Cu which have equal energy.

Fill your ans as sum of digit till you get single integer.

5. Find the total number of electrons of Cr atom which have greater value of Z_{eff} as compared to 3d-electrons.

Fill your ans as sum of digit till you get single integer.

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SECTION-I (i) : (Maximum Marks : 21)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases

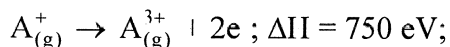
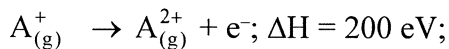
- Select the **correct** order of size :
 (A) $\text{Li} < \text{Ca} < \text{K}$ ~~(B) $\text{Li} < \text{K} < \text{Ca}$~~ (C) $\text{Ca} < \text{Li} < \text{K}$ (D) $\text{K} < \text{Ca} < \text{Li}$
- Which among the following is bad conductor of electricity.
 (A) Iron (B) Nickel ~~(C) Diamond~~ (D) Gold
- Which among the following option is NOT correctly matched :
 (A) Li – Highest metal
 (B) He – Highest ionization energy
~~(C) F – Smallest metal~~ (D) Cs – Most electropositive alkali metal
- IE_1 of Na metal is 5.14 eV/atom. The IE_2 of Na metal is :
 (A) 10.2 eV ~~(B) 47.3 eV/atom~~ (C) 9.8 eV/atom (D) 3.2 eV/atom
- Which among the following representation of electronic configuration do not follow Aufbau's Principle as well as Hund's Rule :

2s	2p	2s	2p
(A) 1	↑↓ 1 1 1	(B) ↑↓	↑↓ 1 1 1 ✗
(C) ↑↓	1 ↓ 1 1 ✗	(D) 1	↑↓ 1 ↓
- All isotopes of a given element have the same _____ :
~~(A) number of protons~~ (B) number of neutrons
 (C) number of electrons and neutrons (D) mass
- The increasing order of atomic radii of the following group 13 elements is
 (A) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ ~~(B) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$~~
 (C) $\text{Al} < \text{In} < \text{Ga} < \text{Tl}$ (D) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$

(ii) : (Maximum Marks : 20)

- This section contains **FIVE** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 In all other cases

8. Consider the following ionization steps :



Which among the following options is/are **CORRECT** :

(A) IE_1 of $A_{(g)}$ is 200 eV/atom

(B) IE_3 of $A_{(g)}^+$ is 550 eV/atom

(C) IE_1 of $A_{(g)}^+$ is 200 eV/atom

(D) IE_3 of $A_{(g)}$ is 550 eV/atom

9. Which of the following compound(s) is/are found in his maximum covalency of central atom.

(A) XeO_3

(B) SF_4

(C) OSF_4

(D) XeO_3F_2

10. In which of the following all atoms which is/are bonded with central atom have covalency = 2

(A) CS_2

(B) XeO_4

(C) $POCl_3$

(D) H_3PO_3

11. Select pair in which same type of covalent bond (w.r.t = σ , π , δ) by overlapping- (X-axis is overlapping axis)

(A) $p_x + p_x$ and $p_y + p_y$

(B) $p_y + p_y$ and $p_z + p_z$

(C) $s + p_x$ and $p_x + p_x$

(D) $d_{xy} + d_{xy}$ and $p_y + d_{xy}$

12. The compound(s) with TWO lone pairs of electrons on the central atom is(are)

(A) BrF_5

(B) ClF_3

(C) XeF_4

(D) SF_4

(iii) Paragraph Type

This section contains **ONE paragraphs** each describing theory, experiment, data etc. **Two questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Passage (For Q.No.13 & 14)

Elements are placed in long form of periodic table in the increasing order of their atomic number. Elements belonging to the same group have similar outer most electronic configuration. A period always start with s-block and ends with p-block.

For given questions assume that value of Azimuthal quantum number varies from 0 to (n + 2).

13. Maximum number of elements which can be arranged in 2nd period in periodic table are :

(A) 8 (B) 18 (C) 36 (D) 2

14. The period number of sulphur atom is :

(A) 1 (B) 2 (C) 3 (D) 4

SECTION-II : Matrix-Match Type

This Section contains **TWO questions**. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the following column :

Column-I

(A) XeF₄ (B) ClF₃ (C) SF₄ (D) NH₃

Column-II

(P) One lone pair on central atom
(Q) Two lone pair on central atom
(R) Three lone pair on central atom
(S) d-orbital in hybridisation
(T) Non planer

2. Match the following column :

Column-I

(Hybridization)

(A) sp³ (B) sp³d (C) sp³d² (D) sp²

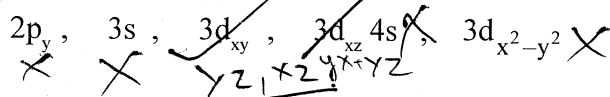
Column-II

(Bond angle)
(P) All adjacent bond angle identical
(Q) Bond angle = 90°
(R) Bond angle = 120°
(S) Bond angle = 180°
(T) Bond angle = 72°

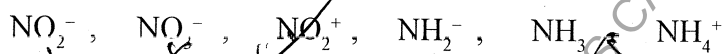
SECTION-IV : (Maximum Marks : 32)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

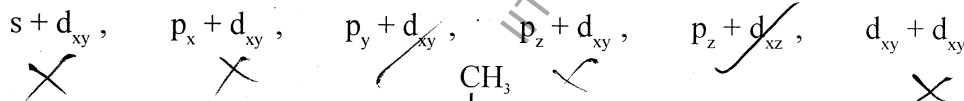
1. How many orbitals among the following have at least one nodal plane in which probability of finding electrons of p_x orbital is zero. **(2)**



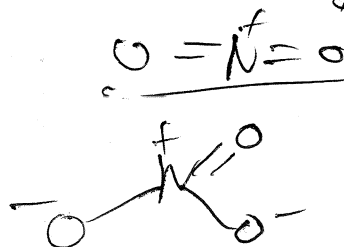
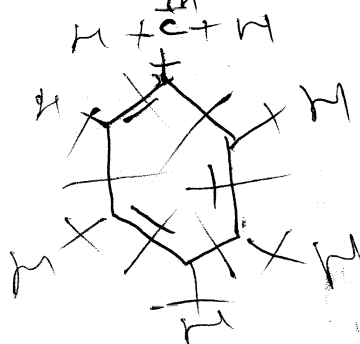
2. Find total number of hybrid orbitals involved in bond formation in Si_2H_6 . **(2)**
3. Find total number of molecules(s)/ion(s) in bond angle of 180° is/are present. **(1)**



4. Find total number of the pair of orbitals which produce π -bond (X-axis is overlapping axis) **(2)**

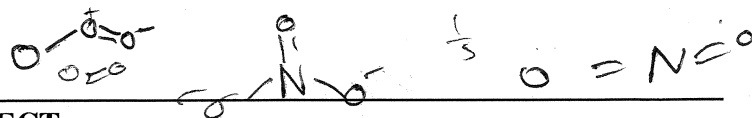


5. Find number of σ -bonds in toluene **(6)**



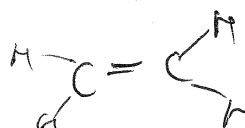
SECTION-I (i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases



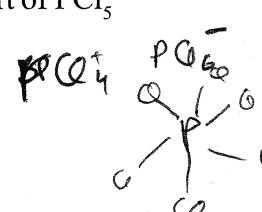
1. Which of the following order is/are **CORRECT** :-

- (A) $\text{CO}_3^{2-} < \text{CO}_2$ (C-O bond length) ☒ (B) $\text{NO}_3^- < \text{NO}_2$ (N-O bond length) ☒
 (C) $\text{O}_2 < \text{O}_3$ (O-O bond length) ☒ (D) $\text{C}_6\text{H}_6 < \text{C}_2\text{H}_4$ (C-C bond length) ☒



2. The number of non-axial set of 'd' orbital participate in the hybridisation of the anionic part of PCl_5

- (A) 1 (B) 2 (C) 3 (D) 0



3. Which of the following statement is **CORRECT** :-

- (A) 'Ne' belongs to s-block ☒ (B) 'La' belongs to f-block ☒
 (C) 'Gd' belongs to d-block ☒ (D) 'F' belongs to p-block ☒

4. Which of the following having second highest bond dissociation energy (out of given molecules).

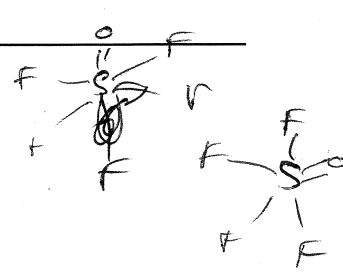
- (A) F_2 (B) Cl_2 (C) Br_2 (D) I_2

(ii) : (Maximum Marks : 16)

- This section contains **FOUR** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 In all other cases

5. Which of the following statement is/are **CORRECT** regarding to SOF_4

- (A) Nodal plane of the π -bond is the molecular plane ☒
 (B) $p\pi-d\pi$ type of π bond is present ☒
 (C) Maximum number of atom in one plane is 4 ☒
 (D) Axial S-F length is greater than equatorial S-F length ☒



6. Which of the following properties changed if all the fluorine atoms from SF_6 is replaced by oxygen atoms without changing the oxidation state of sulphur atom.

- (A) hybridisation of central atom ☒ (B) Dipole moment ☒
 (C) Total number of lone pair ☒ (D) Number of π -bond ☒

7. Which of the following order is/are correct.

- (A) $\text{O}_2\text{SF}_2 < \text{O}_2\text{SCl}_2$ (X-S-X bond angle) ☒ (B) $\text{O}_2\text{SF}_2 > \text{O}_2\text{SCl}_2$ (O-S-O bond angle) ☒
 (C) $\text{O}_2\text{SF}_2 < \text{O}_2\text{SCl}_2$ (Total number of lone pair) ☒ (D) $\text{O}_2\text{SF}_2 > \text{O}_2\text{SCl}_2$ (S-X bond length) ☒

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8. Which of the following molecule/ions have both $p_\pi-p_\pi$ and $p_\pi-d_\pi$ type covalent bond ?

- (A) SO_2Cl_2 ✓ (B) SO_4^{-2} ✗ (C) SO_2 ✗ (D) SO_3 ✗

(iii) Paragraph Type

This section contains **THREE paragraphs** each describing theory, experiment, data etc. **Two questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **ONE OR MORE THAN ONE** correct answers among the four choices (A), (B), (C) and (D). **3(0)**

Passage for Q.No. 09 to 10

According to Gillespie and Nyholm, central atom of valence shell $e^\ominus$ pairs arrange them in space in such direction which cause minimum repulsion between them. Because minimum repulsion = minimum potential energy = maximum stability.

Strength of repulsion = $l.p - l.p > l.p - B.p. > B.p. - B.p.$

9. Which of the following set of molecules/ion have same shape ?

- (A) $\text{SF}_4, \text{BrF}_3$ ✗ (B) $\text{XeOF}_4, \text{XeF}_5^\oplus$ ✗ (C) $\text{ICl}_4^\ominus, \text{TeCl}_4$ ✓ (D) $\text{XeF}_4, \text{IF}_5$ ✓

10. In sp^3d hybridisation which of the following geometry of the molecule is not possible ?

- (A) See-saw ✓ (B) T-shape ✗ (C) Linear ✓ (D) Distorted octahedron ✓

Passage for Q.No. 11 to 12

The attractive force that binds hydrogen atom of one molecule with electronegative atom of the other molecule of the same or different substance is known as hydrogen bond.

11. Which of the following chemical species having inter molecular hydrogen bonding.

- (A) Ortho boric acid ✗ (B) Ortho nitrophenol ✓
(C) Ortho hydroxybenzaldehyde ✗ (D) Ortho chlorophenol ✗

12. Which of the following boiling point order is INCORRECT.

- (A) $\text{H}_3\text{BO}_3 > \text{B(OMe)}_3$ (B) $\text{C}_2\text{H}_5\text{OH} > \text{CH}_3\text{OCH}_3$ ✓
(C) $\text{HF} > \text{HCl}$ ✗ (D) $\text{NH}_3 > \text{SbH}_3$ ✓

Passage for Q.No. 13 to 14

Ionisation energy and electron affinity are two properties on the basis of which many properties can be explained. Ionisation energy is the amount of energy supplied to remove the outermost electrons from an isolated gaseous atom. Electron affinity is the amount of energy when an incoming approaches an isolated gaseous atom.

Column - I Elements	Column - II (IE_1) in (kJ/mol)	Column - III (EA_1) in (kJ/mol)
(I) O(g)	(i) 999	(P) 328
(II) F(g)	(ii) 1251	(Q) 349
(III) Cl(g)	(iii) 1313	(R) 200
(IV) S(g)	(iv) 1681	(S) 141

13. Which of the following is a correct match :

- (A) (II), (iv), (P) ✓ (B) (III), (ii), (Q) ✗ (C) (III), (i), (R) ✗ (D) (IV), (iii), (S) ✗

14. The sum of ($\text{IE}_1 + \text{EA}_1$) for elements in Column I is in order :-

- (A) $\text{Cl} > \text{F} > \text{O} > \text{S}$ ✗ (B) $\text{O} > \text{Cl} > \text{F} > \text{S}$ ✓ (C) $\text{F} > \text{Cl} > \text{O} > \text{S}$ ✗ (D) $\text{F} > \text{O} > \text{Cl} > \text{S}$ ✗

SECTION-II : (Maximum Marks : 16)

- This section contains **TWO** questions.

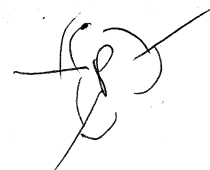
- Marking scheme :

For each entry in **Column-I**

+2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

0 In none of the bubbles is darkened

-1 In all other cases



1. Column-I

(Molecules)

(A) PCl_3F_2

(B) PCl_2F_3

(C) SF_6

(D) IF_7

Handwritten notes for Column-I molecules:
 (A) P S T
 (B) R S T
 (C) P O S T
 (D) Q S T

Column-II

(Characteristics)

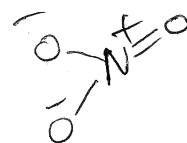
(P) All A-F bonds are equal [A is central atom]

(Q) Number of 90° angle greater than 6

(R) Molecule is polar

(S) Molecule is non-planar

(T) No lone pair on central atom



2. Match the column

Column I

(Compounds)

(A) NO_3^-

(B) XeO_3

(C) SNF_3

(D) ICl_2^+

Handwritten notes for Column I compounds:
 (A) R
 (B) P Q R S T
 (C) P R T
 (D) R T

Column II

(Properties)

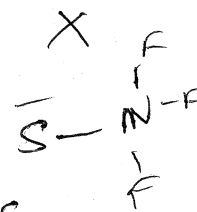
(P) It is non-planar

(Q) It has non zero value of dipole moment

(R) All possible angles are identical

(S) It has pyramidal geometry

(T) It has $d_\pi-p_\pi$ bonding



SECTION-IV : (Maximum Marks : 20)

- This section contains **FIVE** questions.

- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive

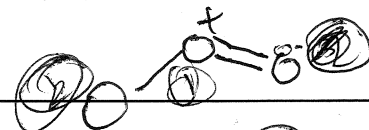
- For each question, darken the bubble corresponding to the correct integer in the ORS

- Marking scheme :

+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases



1. Find the number of species which are planar?

BrF_4^+ , XeOF_2 , BrF_3 , H_2O , H_2S , SF_2

Handwritten answer: 5

2. Find the number of molecule/ions which are polar in nature?

BrF_3 , I_3^- , ICl_4^+ , XeO_3F_2 , PCl_2F_3 , $\text{P}(\text{CH}_3)_3\text{F}_2$

Handwritten answer: 2

3. Find the total number of unshared electron pair(s) in O_3 .

Handwritten answer: 5

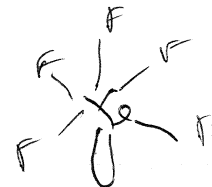
4. Find the maximum number of chemical species which are planar and polar.

HCN , SO_2 , HCl , H_2S , SnCl_2

Handwritten answer: 4

5. Find the maximum number of plane having maximum number of atoms in XeF_5^+ is.

Handwritten answer: 1

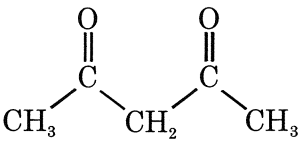
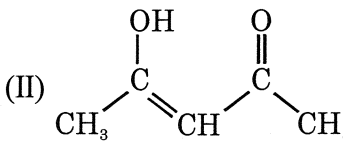
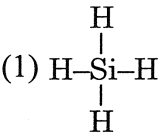
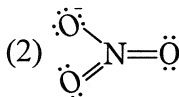
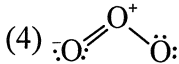


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INORGANIC CHEMISTRY

TIME :

- Select **CORRECT** for H_2CSF_4 molecule :-
(1) All S-F length are identical
(2) It has no π -bond
(3) Four type of bond length data are observed
(4) Only two bonds are at 120°
- Which of the following azimuthal quantum number is not observed for any electron in any 4th period element :-
(1) $l = 0$ (2) $l = 1$ (3) $l = 2$ (4) $l = 3$
- Pair of isotopes is :-
(1) C^{14} , N^{14} (2) O^{16} , O^{18}
(3) O, S (4) ${}^1_1\text{H}^3$, ${}^3_3\text{Li}^7$
- Which is most stable electronic configuration of atom (x) :-
(1) $1s^2 2s^2 2p^2$ (2) $1s^2 2s^1 2p^3$
(3) $1s^2 2s^2 2p^1 3s^1$ (4) All are incorrect
- In which of the following all possible F-P-F bond angles are not identical :-
(1) PCl_3F_2 (2) PCl_2F_3 (3) PF_3 (4) POF_3
- In which of the following at least one double bond is arising from the overlap of unhybridised p-orbitals of both adjacent atoms :-
(1) SO_2Cl_2 (2) SOCl_2
(3) SO_3 (4) All of the above
- (I)  and
(II)  are the two isomer of acetyl acetone. Which have hydrogen bonding :-
(1) I (2) II
(3) I and II both (4) None
- Ionization energy is highest for :-
(1) Zn (2) Cu (3) Ga (4) Sc
- IF_5 is isostructural with :-
(1) PF_5 and BrF_5 (2) XeF_5^+ and XeF_5^-
(3) XeF_5^+ and PF_5 (4) BrF_5 and XeF_5^+
- d-orbital which is/are NOT involve in hybridisation in sp^3d^3 hybridisation :-
(1) d_{xy} (2) d_{yz} (3) $d_{x^2-y^2}$ (4) d_{z^2}
- Select **CORRECT** lewis structure :-
(1)  (2) 
(3) $\text{H}-\ddot{\text{N}}\equiv\text{C}:$ (4) 
- Which of the following pairs of d-orbitals will have electron density along the axes ?
(1) d_{z^2} , $d_{x^2-y^2}$ (2) d_{xy} , $d_{x^2-y^2}$
(3) d_{z^2} , d_{xz} (4) d_{xz} , d_{yz}
- The correct shape and hybridization for XeF_4 are :-
(1) Planar triangle, sp^3d^3
(2) square planar, sp^3d^2
(3) octahedral, sp^3d^2
(4) trigonal bipyramidal, sp^3d
- Among the following which one is a wrong statement?
(1) SeF_4 and CH_4 have same shape
(2) I_3^+ has bent geometry
(3) PH_5 and BiCl_5 do not exist
(4) $p\pi-d\pi$ bonds are present in SO_2
- Which one of the following compounds shows the presence of intramolecular hydrogen bond ?
(1) Cellulose
(2) Concentrated acetic acid
(3) H_2O_2
(4) HCN

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16. How many electrons can fit in the orbital for which $n = 3$ and $l = 1$?
 (1) 10 (2) 14 (3) 2 (4) 6
17. Boric acid is an acid because its molecule
 (1) accepts OH^- from water releasing proton
 (2) combines with proton from water molecule
 (3) contains replaceable H^+ ion
 (4) gives up a proton
18. The hybridizations of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are
 (1) sp , sp^2 and sp^3 (2) sp^2 , sp and sp^3
 (3) sp , sp^3 and sp^2 (4) sp^2 , sp^3 and sp
19. Which of the following fluoro-compounds is most likely to behave as a Lewis base ?
 (1) CF_4 (2) SiF_4 (3) BF_3 (4) PF_3
20. Which of the following pairs of ions is isoelectronic and isostructural ?
 (1) SO_3^{2-} , NO_3^- (2) ClO_3^- , SO_3^{2-}
 (3) CO_3^{2-} , NO_3^- (4) ClO_3^- , CO_3^{2-}
21. Identify the cation having minimum magnetic moment.
 (1) Ni^{+2} (2) Cr^{+1} (3) Fe^{+2} (4) Mn^{+3}
22. The maximum number of orbitals associated to 3rd shell.
 (1) 9 (2) 18
 (3) 7 (4) None of these
23. Oxygen does not form OF_6 because :-
 (1) Due to non availability of vacant 'd' orbitals in 'O'
 (2) Oxygen is second most electronegative element
 (3) OF_6 does not follow octet rule
 (4) None of these
24. Which of the following molecule having highest bond strength :-
 (1) HF (2) HCl (3) HBr (4) HI
25. Which of the following does not show variable covalence :-
 (1) F (2) Cl (3) Br (4) I
26. What is the quantum number value is possible of electron associated to outer shell of Cu in their ground state :-
- | | n | l | m | s |
|-----|---|---|----|----------------|
| (1) | 4 | 1 | 0 | $+\frac{1}{2}$ |
| (2) | 3 | 2 | -1 | $-\frac{1}{2}$ |
| (3) | 4 | 0 | 0 | $-\frac{1}{2}$ |
| (4) | 3 | 1 | -1 | $-\frac{1}{2}$ |
27. Find the electron gain enthalpy of $\text{F}_{(g)}$ using the following data. All the values are in KJ/mole at 25°C.
 $\Delta H_{\text{diss}}(\text{F}_2) = 160 \text{ KJ/mole}$;
 ΔH_f and $\text{NaF(s)} = -571 \text{ KJ/mole}$
 IE_A of $\text{Na(g)} = 494 \text{ KJ/mole}$, $\Delta H_{\text{sub.}}$ of $\text{Na(s)} = 101 \text{ KJ/mole}$, $\text{LE of NaF} = -352 \text{ KJ/mole}$
 (1) -571 (2) -671 (3) -894 (4) -1495
28. Which of the following orbital(s) can NOT form π -bond if z-axis is the approaching axis :-
 (1) $p_x - p_x$ (2) $d_{yz} - d_{yz}$
 (3) $d_{xz} - d_{xz}$ (4) $d_{xz} - p_z$
29. Which of the following Lewis acid, Lewis base interaction is NOT possible :-
 (1) $\text{BeF}_2 + 4\text{F}^- \rightarrow \text{BeF}_6^{4-}$
 (2) $\text{SiF}_4 + 2\text{F}^- \rightarrow \text{SiF}_6^{2-}$
 (3) $\text{NH}_3 + \text{BF}_3 \rightarrow \text{H}_3\text{N} \cdot \text{BF}_3$
 (4) $\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{H}_3\text{O}^+$
30. In ICl_2^+ , ICl_2^- and ICl_4^- sum of the bond pairs and lone pairs on each iodine atom are respectively :
 (1) 4, 5 and 6 (2) 2, 2, 4
 (3) 4, 5 and 4 (4) 2, 3 and 2

INORGANIC CHEMISTRY

TIME :

SECTION-I (i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. In which of the following inter particle interaction is strongest ?
(A) Between Xe & H₂O (B) Between KF & HF
(C) Liquid He (D) Between KI & I₂
2. Among the following mixtures, dipole-dipole as the major interaction, is present in
(A) KCl and water (B) benzene and carbon tetrachloride
(C) benzene and ethanol (D) acetonitrile and acetone
3. Which of the following has **CORRECT** order of bond length of their indicated bond(s) ?
(A) d_{O-O} ; H₂O₂ < O₂F₂ (B) d_{C-O} ; CO₃²⁻ > CO₂ > CO
(C) d_{O-O} ; O₃ < O₂ (D) d_{C-C} ; H₃C-CH₃ < CF₃-CF₃
4. Which of the following statement is **INCORRECT** about PCl₃F₂
(A) Molecule is non polar
(B) P-F bond length is greater than P-Cl bond length
(C) Axial hybrid orbital length is greater than that of equatorial hybrid orbital length
(D) All F - P - Cl bond angles are equal
5. Which of the following order is **CORRECT** :-
(A) N₂H₂ < N₂F₂ → N-N bond length (B) H₂O₂ < O₂F₂ → O-O bond length
(C) O₂F₂ > OF₂ → O-F bond length (D) B₂F₄ > B₂Cl₄ → B-B bond length

(ii) : (Maximum Marks : 20)

- This section contains **FIVE** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 In all other cases

6. In which of the following bond angle difference is greater than 10° :
(A) NH₃, PH₃ (B) BF₃, CF₄ (C) PH₃, AsH₃ (D) NH₃, H₂O

7. Select option(s) in which importance of London dispersion forces is increased on moving from left to right
 (A) $\text{CH}_{4(l)}$, $\text{NH}_{3(l)}$, $\text{H}_2\text{O}_{(l)}$ (B) $\text{CH}_{4(l)}$, $\text{SiH}_{4(l)}$, $\text{GeH}_{4(l)}$
 (C) $\text{Xe}_{(l)}$, $\text{Kr}_{(l)}$, $\text{Ar}_{(l)}$ (D) $\text{F}_{2(l)}$, $\text{Cl}_{2(l)}$, $\text{Br}_{2(l)}$
8. Which of the following statements are **CORRECT** for ClF_3 molecules
 (A) Number of bond pair – lone pair repulsions at central atom are '6'
 (B) Axial d orbital is involved in hybridization of central atom
 (C) Maximum two Cl–F bonds are identical
 (D) Total number of lone pairs in molecule are '11'
9. Consider the CH_2SF_4 molecule choose the **CORRECT** statements.
 (A) There are two d-orbitals use in bonding
 (B) Nodal-plane of π -bond in axial plane
 (C) Maximum number of atom in a plane are six
 (D) Hydrogen atom are in equatorial plane
10. Which of the following pair of species both posses two lone pairs of electrons on central atom and planar structure.
 (A) SF_4 & XeF_4 (B) XeF_4 & $\text{ICl}_4^\ominus$ (C) ClF_3 & H_2O (D) $\text{XeF}_5^\ominus$ & $\text{I}_3^\oplus$

SECTION-I(iii) : (Maximum Marks : 9)

- This section contains **ONE** paragraph.
- Based on each paragraph, there will be **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
 +3 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
 0 In all other cases

Paragraph for Question 11 to 13

Valence bond theory is one of the two quantum mechanical approaches that explains bonding in molecules as well explains the geometry of the molecules.

11. The hybridisation of atomic orbitals of nitrogen atom in $\text{NO}_2^\oplus$, $\text{NO}_3^\ominus$ and $\text{NH}_4^\oplus$ are :
 (A) sp^2 , sp^3 & sp^3 (B) sp , sp^2 & sp^3 (C) sp^2 , sp^2 & sp^3 (D) sp , sp^3 & sp^3d
12. The order of increasing N–O bond length in $\text{NO}_3^\ominus$, $\text{NO}_2^\ominus$, $\text{NO}_2^\oplus$ is :
 (A) $\text{NO}_2^\oplus < \text{NO}_2^\ominus < \text{NO}_3^\ominus$ (B) $\text{NO}_2^\ominus < \text{NO}_2^\oplus < \text{NO}_3^\ominus$
 (C) $\text{NO}_3^\ominus < \text{NO}_2^\ominus < \text{NO}_2^\oplus$ (D) $\text{NO}_2^\oplus < \text{NO}_3^\ominus < \text{NO}_2^\ominus$
13. When BF_3 is mixed with NH_3 in 1 : 1 ratio then the correct statement about product is :
 (A) Number of bonds in N & B remains unchanged
 (B) The hybridisation state of B is changed from sp^2 to sp^3
 (C) The coordinate or dative bond is not present in the product
 (D) Nitrogen atom contains one lone pair in the product

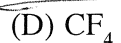
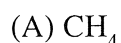
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SECTION-II : Matrix-Match Type

This Section contains **ONE** question. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the following column :

Column-I

(Molecules)



Column-II

(Characteristics given among the molecules of column-I)

(P) Molecule is having perfect tetrahedral shape

(Q) C-F bond has maximum p-character

(R) C-H bond has maximum s-character

(S) Molecule is having maximum number of equal angles

(T) Molecules has lowest bond angle

SECTION-IV : (Maximum Marks : 20)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. Find out the total number of **CORRECT** orders of indicated properties :

(I) C—C > Si—Si : Bond dissociation energy

(II) Cl—Cl > F—F : Bond dissociation energy

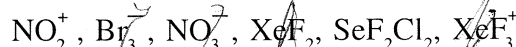
(III) O = O > S = S : π -Bond strength

(IV) PH₃ > NH₃ : Bond energy of E—H bond (where E = central atom)

(V) H—F > F—F : Thermal stability

(VI) 2p—2s > 2p—2p > 2s—2s : Bond strength

2. Find the number of molecule which are planar but central atom is not sp² hybridised



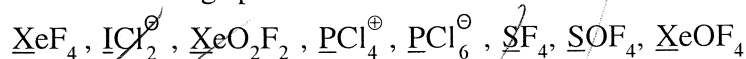
3. Find the number of species which have 2p π —3d π bond(s) in their most stable structure



4. The number of σ -bonds in P₄O₁₀ is :

Fill your answer as sum of digits till you get the single digit answer.

5. The number of species which consists of sp³d hybridised central atom for the underlined atoms with in the following species / molecules is / are :



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INORGANIC CHEMISTRY

TIME :

SECTION-I (i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
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0 If none of the bubbles is darkened
-1 In all other cases

- ✓ 1. Dipole-induced dipole interaction depends upon
(A) size of polarisable particle (B) dipole moment of permanent dipole
(C) Both (A) and (B) (D) None of these
2. Which of the following does not have perfect tetrahedral geometry.
(A) NH_4^+ (B) BF_4^- (C) CH_2Cl_2 (D) ClO_4^-
3. Which of the following statement is **INCORRECT** about PCl_3F_2 .
(A) Molecule is non polar
(B) P-F bond length is greater than P-Cl bond length
(C) Axial hybrid orbital length is greater than that of equatorial hybrid orbital length
(D) All F-P-Cl bond angles are equal
4. What is the hybridisation of the anionic part of $\text{Cl}_2\text{O}_6(\text{s})$:-
(A) sp^3 (B) sp^3d (C) sp^3d^2 (D) sp^2

SECTION-I (ii) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has FOUR options (A), (B), (C) and (D). **One or more than one** option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

5. Which of the following statement is/are CORRECT for BCl_3 :-
(A) One unhybridised 2p orbital is on B (B) Act as lewis acid
(C) All bond angles are equal (D) Boron is sp^2 hybridised

6. Identify the chemical species in which all A-O bond lengths are equal (where A is underlined atom)

- (A) $\underline{\text{C}}\text{O}_3^{2-}$ (B) $\underline{\text{H}}\text{CO}_3^-$ (C) $\underline{\text{N}}\text{O}_2^-$ (D) $\underline{\text{S}}\text{O}_4^{2-}$

7. In which of the following σ -bond(s) is/are formed by 4p-1s overlapping of two atoms

- (A) GeH_4 (B) AsH_3 (C) H_2Se (D) HCl

8. Which of the following is/are planar :-

- (A) I_3^+ (B) I_3^- (C) IF_3 (D) ICl_4^-

9. Which of the following option(s) is/are **CORRECT** ?

Type of interaction distance-energy function

- (A) dipole-dipole $\frac{1}{r^3}$ (B) H-bonding $\frac{1}{r^3}$

- (C) Ion-dipole $\frac{1}{r^2}$ (D) London force $\frac{1}{r^6}$

10. Which of the following molecules have **CORRECT** indicated overlapping.

Molecule Overlapping in the bridge bond (if any)

- (A) Si_2Cl_6 $\text{sp}^3\text{-p-sp}^3$
(B) Be_2H_4 $\text{sp}^2\text{-s-sp}^2$
(C) Si_2H_6 $\text{sp}^3\text{-s-sp}^3$
(D) B_2H_6 $\text{sp}^3\text{-s-sp}^3$

SECTION-I(iii) : (Maximum Marks : 9)

- This section contains **ONE** paragraphs.
- Based on each paragraph, there will be **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

+3 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened

0 In all other cases

Paragraph for Q.No. 11 and 12

It is defined as the angle between the orbitals containing bonding electron pairs around the central atom in a molecule/complex ion. Bond angle is expressed in degree which can be experimentally determined by spectroscopic methods.

11. What is the bond angle of NO_2^- :-

- (A) 115° (B) 120° (C) 134° (D) 107°

12. Which of the following order of bond angle is **INCORRECT** :-

- (A) $\text{PH}_3 < \text{PH}_4^+$ (B) $\text{NH}_3 < \text{NH}_4^+$ (C) $\text{CH}_3^- < \text{CH}_3^+$ (D) $\text{NH}_3 < \text{NF}_3$

Paragraph for Q.No. 13 and 14

Bent's rule can be stated as follows. "The central atom projects the hybrid orbitals of less s-character" towards more electronegative atom.

13. Which of the following statement is **CORRECT** ?

- (A) CH_3F is not perfect tetrahedral
(B) PCl_3F_2 has got a trigonal bipyramidal (T.B.P) shape.
(C) In PCl_3F_2 , the two fluorine atoms preferably are positioned in the axial directions.
(D) All are correct

14. Which of the following order is **CORRECT** ?

- (A) $d_{\text{C-H}}$ in $\text{CH}_3\text{Cl} > d_{\text{C-H}}$ in CH_3F
(B) $d_{\text{C-H}}$ in $\text{CH}_3\text{Cl} < d_{\text{C-H}}$ in CH_3F
(C) $\widehat{\text{H}}\text{P}\widehat{\text{H}}(\text{PH}_3) > \widehat{\text{H}}\text{P}\widehat{\text{H}}(\text{PH}_4^+)$
(D) $\text{H}-\widehat{\text{C}}-\text{H}$ in $\text{CH}_4 > \text{F}-\widehat{\text{C}}-\text{F}$ in CF_4

Paragraph for Q.No. 15 and 16

Back bonding is a type of sideways overlapping.

15. Which of the following molecule has $2p_\pi - 3d_\pi$ back bonding.

- (A) PCl_3 (B) CCl_3^- (C) BCl_3 (D) $(\text{BO}_2^-)_3$

16. Which of the following has **CORRECT** order of strength of back bonding.

- (A) $\text{BF}_3 > \text{BCl}_3$ (B) $\text{O}(\text{SiH}_3)_2 > \text{S}(\text{SiH}_3)_2$
(C) $\text{N}(\text{SiH}_3)_3 < \text{NH}_2\text{SiH}_3$ (D) All are correct

SECTION-II : Matrix-Match Type

This Section contains **ONE** questions. Each question has **four** statements (A, B, C and D) given in **Column I** and **five** statements (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

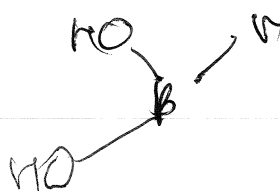
8(0)

1. **Column-I**

- (A) BF_3 R S T P
(B) BCl_3 R S T P Q
(C) H_3BO_3
(D) $\text{B}(\text{OCH}_3)_3$ P R T S

Column-II

- (P) Intra molecular lewis acid-base interaction
(Q) $2p_\pi - 3p_\pi$ back bond
(R) Lewis acid
(S) Incomplete octet of central atom boron
(T) sp^2 hybridisation of B



SECTION-IV : (Maximum Marks : 12)

- This section contains **THREE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. Find the maximum number of identical bond angle in ClO_3^- .

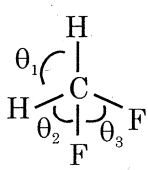
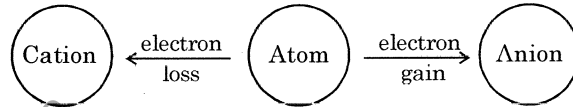
2. Find the number of chemical species having $p\pi-p\pi$ bond.

ClO_2^- ; NO_2^- ; CO ; CO_3^{2-} ; PO_4^{3-} ; XeO_3

3. Find the number of molecules in which $\text{F}-\hat{\text{C}}-\text{F}$ bond angle of COF_2 is higher than that of given molecules ($\text{X}-\hat{\text{A}}-\text{X}$), where ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{H}$).

BCl_3 , BF_3 , COCl_2 , COH_2 , COBr_2

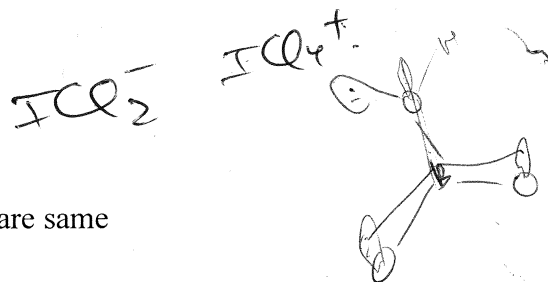
- Which of the following is a valid set of four quantum numbers ? (n, l, m_l, m_s)
(1) 2, 1, +2, +1/2 ☒ (2) 2, 1, 0, +1/2
(3) 1, 1, 0, -1/2 (4) 2, 2, 1, -1/2
- Select the process which is exothermic ?
(1) $\text{Cl}_{(g)} \longrightarrow \text{Cl}_{(g)}^+ + e^-$
(2) $\text{O}_{(g)}^- + e^- \longrightarrow \text{O}_{(g)}^{2-}$ ☒
(3) $\text{Ar} - e^- \longrightarrow \text{Ar}^+$
(4) $\text{N}_{(g)} \longrightarrow \text{N}_{(g)} + e^-$
- The hybridization of Br atom in Br_3^- is the same as :-
(1) Xe in XeO_3 (2) I in ICl_4^-
(3) Xe in XeO_2F_2 (4) S in SF_6
- Which of the following molecule is polar in nature ?
(1) BF_4^- (2) CF_4
(3) SF_4 (4) XeF_4
- Which molecule has maximum number of S-S bonds ?
(1) $\text{H}_2\text{S}_3\text{O}_6$ (2) S_3O_9
(3) $\text{H}_2\text{S}_2\text{O}_3$ (4) S_8
- Which of the following molecules/ions are hypovalent :-
(1) CO (B) Al_2Cl_6
(3) BF_3 (D) AlF_3
- Species in which central atom has sp hybridization :-
(1) HF ☒ (B) COS ☒
(3) XeF_2 (D) NO_2^-
- Which of the following orders are **CORRECT**
(1) $sp-sp > sp-sp^2 > sp^2-sp^2 \rightarrow$ Strength of overlapping
(2) $2p_\pi-3p_\pi > 2p_\pi-3d_\pi \rightarrow$ Strength of overlapping
(3) $2s-2s > 2s-2p > 2p-2p \rightarrow$ Strength of overlapping
(4) None of these
- Which of the following molecule/ion has sp^3 hybridisation :-
(1) SF_4 (2) TeCl_4
(3) PH_4^+ (4) None of these
- Species in which π -coordinate bond is present :-
(1) NO_3^- (2) POCl_3
(3) CO (4) NOF_3
- Species in which $p_\pi-d_\pi$ bonding is present.
(1) CO_2 (2) SNF_3
(3) SiH_4 (4) CO_3^{2-}
- Which of the following order is **CORRECT** :-
(1) $\text{N}_2\text{H}_2 < \text{N}_2\text{F}_2 \rightarrow \text{N-N bond length}$
(2) $\text{H}_2\text{O}_2 < \text{O}_2\text{F}_2 \rightarrow \text{O-O bond length}$
(3) $\text{O}_2\text{F}_2 > \text{OF}_2 \rightarrow \text{O-F bond length}$
(4) $\text{B}_2\text{F}_4 > \text{B}_2\text{Cl}_4 \rightarrow \text{B-X bond length}$
- Which of the following atomic orbital have least probability of involvement in hybridization of central atom of given molecules :-
 $\text{CS}_2, \text{PCl}_5, \text{SF}_6, \text{ClF}_3$
(1) d_{xy} (2) $d_{x^2-y^2}$
(3) d_{z^2} (4) p_z
- Which of the following combination is **CORRECT** if internuclear axis is 'z' :-
(1) $s + p_x \rightarrow \sigma\text{-bond}$
(2) $p_x + p_y \rightarrow \pi\text{-bond}$
(3) $d_{xy} + d_{xy} \rightarrow \delta\text{-bond}$
(4) $p_y + d_{yz} \rightarrow \delta\text{-bond}$
- Species in which underlined atom has (-1) formal charge :-
(1) $\underline{\text{C}}\text{N}^-$ (2) $\underline{\text{N}}\text{O}_3^-$
(3) $\underline{\text{P}}\text{Cl}_3$ (4) $\underline{\text{C}}\text{S}_2$

16. Which of the following order is **INCORRECT** :-
 (1) $\text{SiF}_4 = \text{SiCl}_4$ (2) $\text{XeF}_2 = \text{ICl}_2^-$
 (3) $\text{PH}_3 > \text{PF}_3$ (4) $\text{PCl}_3 < \text{PBr}_3$
17. Which of the following molecule have capped octahedral structure :-
 (1) XeF_6 (2) IF_6^-
 (3) Both (1) and (2) (4) None of these
18. For above species the **CORRECT** order of bond angle is :-

 (1) $\theta_1 > \theta_3 > \theta_2$ (2) $\theta_2 > \theta_1 > \theta_3$
 (3) $\theta_1 > \theta_2 > \theta_3$ (4) $\theta_3 > \theta_2 > \theta_1$
19. Which of the following have both $3c-2e^\ominus$ and $2c-2e^\ominus$ type of bond ?
 (1) $\text{BeH}_2(\text{s})$ (2) $\text{BeCl}_2(\text{s})$
 (3) Al_2H_6 (4) Si_2Cl_6
20. Which of the following given mineral is example of sheet silicate ?
 (1) $\text{Na}_2\text{Fe}_3^{II}\text{Fe}_2^{III}[\text{Si}_8\text{O}_{22}](\text{OH})_2$
 (2) $\text{Ca}_3[\text{Si}_3\text{O}_9]$
 (3) $\text{Al}_2(\text{OH})_2[(\text{Si}_2\text{O}_5)_2]$
 (4) $\text{Zn}_4(\text{OH})_2[\text{Si}_2\text{O}_7]$
21. Which of the following ligand is not responsible for symmetrical bond cleavage in B_2H_6 ?
 (1) Me_2S (2) H^-
 (3) THF (4) NH_3
22. Which of the following molecule have $3C-4e^\ominus$ bond as well as planar geometry ?
 (1) Al_2Br_6 (2) Al_2I_6
 (3) I_2Cl_6 (4) $(\text{AlH}_3)_n$
23. The possible electronic configuration of Vanadium atom/ion in the compound, where it produces spin only magnetic moment of 1.73 BM is :-
 (1) $[\text{Ar}]4s^23d^3$ (2) $[\text{Ar}]4s^03d^1$
 (3) $[\text{Ar}]4s^13d^0$ (4) $[\text{Ar}]4s^03d^2$
24. Which of the following is **CORRECT** option for the following statements (T = True, F = False) :-
 (1) Value of IE_2 is maximum for Na amongst 3rd period elements
 (2) Zn is coinage metal
 (3) Al is typical as well as bridge element
 (4) $\text{EGE}_1 > 0$ for inert gases
 (1) TTTF (2) TFTF (3) TFFT (4) FTFT
25. Isobar of 'Ni' element with ${}_{26}\text{Fe}^{58}$ will have neutrons equal to :-
 (1) 28 (2) 30 (3) 32 (4) 31
26. _____ Occupy the same place in the periodic table :-
 (1) Isobars (2) Isotopes
 (3) Isotones (4) None of these
27. 
 The property which is same for both cation and anion of an atom :-
 (1) Size
 (2) Tendency to accept electrons
 (3) Tendency to loose electrons
 (4) None of these
28. Find the total number of σ and π bonds in HCP :-
 (1) $2\sigma, 1\pi$
 (2) $2\sigma, 2\pi$
 (3) $1\sigma, 2\pi$
 (4) Molecule does not exist
29. Choose the metal 'M' for which its oxide will have maximum lattice energy amongst these
 (1) Na (2) Mg
 (3) Al (4) Ba
30. Diagonal relationship is shown by :-
 (1) All elements with their diagonally opposite elements
 (2) All elements of 2nd and 3rd period
 (3) Some elements of 2nd and 3rd period
 (4) Elements of 3rd and 4th period only

SECTION-I (i) : (Only One option correct Type)

- This section contains **TWO** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
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0 If none of the bubbles is darkened
-1 In all other cases

1. Which of the following is correct for $I_2Cl_6(s)$?
- (A) It is non-planar like Al_2Cl_6
(B) It has both $2c-2e^-$ & $3c-2e^-$ bond
(C) In it's molted form cationic part is non-planar
(D) In it's-molted form anionic part all closest bond angles are same
2. In which of the following silicate is double chain silicate :
- (A) $Ca_2Mg_5[(Si_4O_{11})_2](OH)_2$ (B) $Be_3Al_2[Si_6O_{18}]$
(C) $Mg_3(OH)_4[Si_2O_5]$ (D) $Al_2(OH)_4[Si_2O_5]$



SECTION-I (ii) : (One or more options correct Type)

- This section contains **FIVE** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 In all other cases

3. In which of the following compounds bond angle is affected by back bonding :

(A) PF_3 (B) $P(SiH_3)_3$ (C) CCl_3^- (D) $O(CH_3)_2$

4. Which of the following order is correct (d = distance) ?

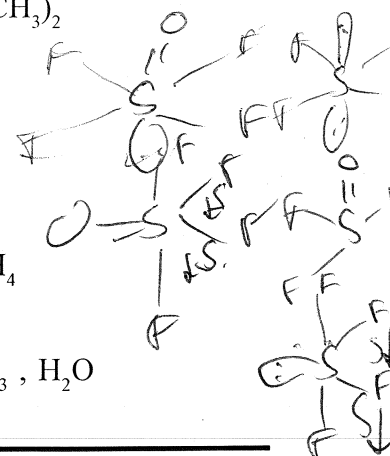
(A) $d_{O-F} \Rightarrow O_2F_2 > OF_2$ (B) $d_{S-F_{eq}} \Rightarrow SOF_4 < SF_4$
(C) $d_{P=O} \Rightarrow POF_3 > POCl_3$ (D) $d_{S-F_{eq}} \Rightarrow SOF_4 > SF_4$

5. Which of the following is/ are planar :

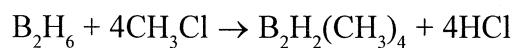
(A) $B(OH)_3$ (B) BO_3^{3-} (C) $B_3N_3H_6$ (D) C_2H_4

6. In which of the following bond angle difference is greater than 10° :

(A) NH_3, PH_3 (B) BF_3, CF_4 (C) PH_3, AsH_3 (D) NH_3, H_2O



7. For following reaction DOWNLOAD FROM IITIANS CAREER



(X) (Y)

Select the correct statement(s) :

- (A) Only terminal hydrogen atoms of (X) are replaced
- (B) In (X) and (Y) number of $3\text{c}-2\text{e}^-$ bonds are same
- (C) In (X) and (Y) number of $2\text{c}-2\text{e}^-$ bonds are different
- (D) Hybridisation of 'B' in (X) and (Y) is same

SECTION-I(iii) : (Paragraph type)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there will be **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

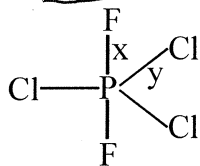
+3 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened

0 In all other cases

PARAGRAPH (8 TO 9)

According to Bent's rule lone pair of electron prefers to occupy that hybrid orbital which has greater percentage of s-character while more electronegative atom / group prefers to occupy that hybrid orbital which has smaller percentage of s-character.

8. Define the correct order of hybrid orbital length of P-atom in given compound :



☒ (A) $x < y$

(B) $x > y$

(C) $x = y$

(D) can't say

9. Define the maximum number of identical bond angle in CH_2Cl_2 :

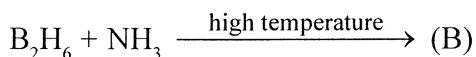
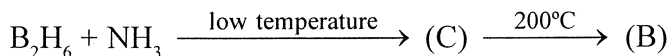
(A) 2

(B) 3

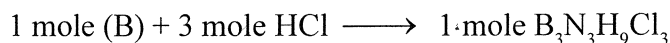
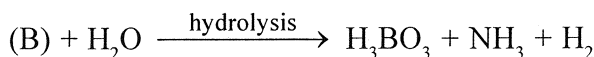
☒ (C) 4

(D) 6

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PARAGRAPH (10 TO 11)



1 : 2



10. When (A) undergoes in hydrolysis then oxy-acid of boron is formed then identify (A) :

(A) SbCl_3

(B) BCl_3

(C) BiOI

(D) BiOCl

11. Select the correct statement about (B)

(A) (B) is known as inorganic benzene

(B) (B) is more reactive than organic benzene due to bond polarity

(C) (B) has $2p_\pi-2p_\pi$ back bonding

(D) All are correct

SECTION-I(iii) : (Matching List Type)

- This section contains **TWO** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- Marking scheme :

+3 If the bubble corresponding to the correct option is darkened

0 In all other cases

12.	List-I Chemical species	List-II Characteristic
	(P) $\text{B}(\text{OH})_3$	(1) $2p_\pi-2p_\pi$ type of back bonding is involved
	(Q) CCl_3^-	(2) $2p_\pi-3d_\pi$ type of back bonding is involved
	(R) $:\text{CCl}_2$ (singlet)	(3) $3p_\pi-2p_\pi$ type of back bonding is involved
	(S) NF_3	(4) No back bonding is involved

Code :

	(P)	(Q)	(R)	(S)
(A)	1	2	3	4
(B)	4	3	2	1
(C)	1	3	4	2
(D)	2	4	3	1

13. List-I DOWNLOADED FROM IITIANSCAREER

Chemical species	Maximum number of atom in one plane
(P) $B_3N_3H_6$	(1) 5
(Q) $I(CN)_2^-$	(2) 12
(R) XeF_5^-	(3) 4
(S) $XeOF_4$	(4) 6

Code :

(P)	(Q)	(R)	(S)	(P)	(Q)	(R)	(S)
(A) 4	3	2	1	(B) 2	1	4	3
(C) 1	3	4	2	(D) 2	4	3	1

SECTION-II : (Matrix-Match Type)

This Section contains **TWO** questions. Each question has **four** statements (A, B, C and D) given in **Column I** and **five** statements (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. 8(0)

1. Column-I

(Species / Ions)

(A) PCl_5

(B) SF_4

(C) BrF_4^-

(D) CH_2SF_4

Column-II

(Correct characteristics)

(P) Axial bond length is longer than equatorial bond length

(Q) d_{z^2} -orbital is utilised in hybridisation of underlined atom

(R) At least one bond angle $\leq 90^\circ$

(S) Underlined atom has lone pair of electrons

(T) Non planar molecule

2. Column-I

(Species / Ions)

(A) $Si_3O_9^{6-}$

(B) S_3O_9

(C) $P_3O_9^{3-}$

(D) $S_3O_6^{2-}$

Column-II

(Correct characteristics)

(P) Cyclic structure

(Q) Have only two tetrahedron

(R) All tetrahedron share two vertex

(S) Central atom in highest O.N.

(T) sp^3 hybridise underline atom

DOWNLOAD FROM IITIANSCAREER
SECTION-IV : (Integer type)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. Find out the total number of **CORRECT** orders of indicated properties :

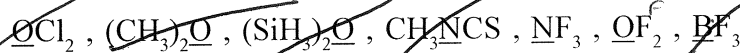
- (I) $2p-2s > 2p-2p > 2s-2s$: Bond energy ✓
 (II) $PH_3 > NH_3$: Bond energy of E-H bond (where E = central atom) ✓
 (III) $O=O > S=S$: Stability ✓
 (IV) $Cl-Cl > F-F$: Bond energy ✓
 (V) $H-F > F-F$: Thermal stability ✓
 (VI) $C-C > Si-Si$: Strength of bond ✓

(4)

2. The number of molecules having bond angle 120° with respect to underlined atom is / are :



3. The total number of following species having larger bond angle than methane (CH_4) with respect to underlined atom.



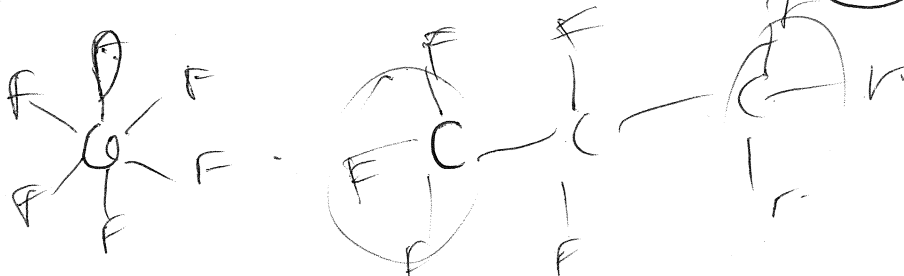
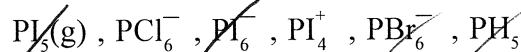
4. Find the total number of species in which the underlined atom is sp^3 hybridized.



5. Define the maximum number of identical C-F bond length in given compound.



6. Find the number of molecules / ions which do not exist due to steric repulsion



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IITIANS CAREER

SECTION-I (I)

- This section contains **10** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

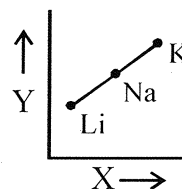
1. If coordinate 'X' is atomic number then select best for 'Y' coordinate :

(A) IE_1

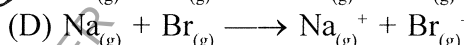
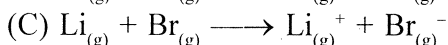
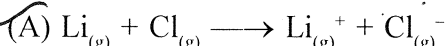
(B) Electron affinity

(C) Z_{eff} (using Slater rule)

(D) $\frac{1}{\text{electronegativity}}$



2. Maximum enthalpy change is associated with



3. Set of species in which 3rd principal level is completely filled

(A) Ar, Zn

(B) Cu, Zn

(C) Cu^{+2} , Zn^{+2}

(D) Cl^- , Ar

4. XeF_4 has total 36 valence electrons. Select correct for uses of valence electrons in XeF_4

(A) 8 valence electrons used to make Xe - F bonds

(B) 24 valence electrons used to make in lone pair of F-atoms

(C) 4 valence electrons used to make in two lone pairs of Xe

(D) All of the above

5. Bond angles 104° , 117° and 180° are associated with

(A) NH_3 , NO_2^- and NO_2^+ respectively

(B) H_2O , NH_3 and CO_2 respectively

(C) H_2O , O_3 and NO_2^- respectively

(D) H_2O , O_3 and CO_2 respectively

6. Identify the **INCORRECT** match(es) in the following :-

(A) XeF_3^+ : sp^3d (Bent T-shape)

(B) XeF_5^- : sp^3d^3 (Pentagonal planar)

(C) XeO_3F_2 : sp^3d^2 (square planar)

(D) XeO_2F_2 : sp^3d (see-saw)

7. What is the **CORRECT** geometry and hybridisation for SF_4 ?

(A) Trigonal bipyramidal, sp^3d

(B) Tetrahedral, sp^3

(C) See-Saw, sp^3d

(D) Square planar, dsp^2

8. Which of the following molecules give oxyacid as well as hydra acid on hydrolysis at room temperature

(A) SF_4

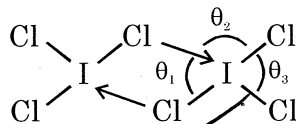
(B) P_4O_{10}

(C) PCl_3

(D) (A) and (C) both

9. Which of the following odd e^- species is planar ?
(A) CH_3^+ (B) ClO_3 (C) CF_3 (D) None of these

10. In the molecule I_2Cl_6 , choose the **CORRECT** statement(s)



- (A) The molecule is planar
(B) 8 number of angles of the type θ_2 are present
(C) $\theta_3 > \theta_1 > \theta_2$
(D) Total 4 unshared lone pairs are present

SECTION - I (II)

- This section contains 7 questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 In all other cases

11. Select set of quantum number which is associated with any electron in Cr ($Z = 24$)

- (A) $n = 4, l = 1, m = 0, s = +\frac{1}{2}$ ✗
(B) $n = 4, l = 0, m = 0, s = -\frac{1}{2}$ ✗
(C) $n = 3, l = 2, m = -2, s = +\frac{1}{2}$ ✗
(D) $n = 3, l = 1, m = +1, s = -\frac{1}{2}$ ✗

12. (I) POF_3 (II) SOF_4 (III) IOF_5

On moving I to III which change is observed

(A) Number of d-orbitals involved in bonding	(I) One → (II) Two → (III) Three
(B) Number of orbitals involved in hybridisation	(I) Four → (II) Five → (III) Six
(C) Number of $d-p_\pi$ bonds in the respective molecule	(I) One → (II) Two → (III) Three
(D) Number of lone pairs on central atom	(I) Zero → (II) Zero → (III) Zero

13. Which of the following order **INCORRECT** between PCl_3F_2 and PCl_2F_3 :-

- (A) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (Dipole moment)
(B) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (Total number of lone pair)
(C) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (Maximum number of atom in one plane)
(D) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (P-Cl Bond length)

14. Diborane is hydride of boron and it exist in gaseous state and poisonous in nature. Which of the following statement is/are **CORRECT** for diborane :-
- (A) It is e^- deficient species (B) Banana bond is present
(C) Boron is sp^3 hybridised (D) Bridge bond is longer than B-H_t (t = terminal)
15. Which of the following properties are **CORRECT** for mineral tremolite $[Ca_2Mg_5(Si_4O_{11})_2(OH)_2]$:
- (A) Double chain silicate unit is present (B) Silicate unit is non-planar
(C) Some tetrahedron units are sharing two corners/O-atoms
(D) Some tetrahedron units are sharing three corners/O-atoms
16. Which of the following order of boiling point is **INCORRECT** :-
- (A) $BF_3 < B(CH_3)_3$ (B) $B(OH)_3 > B(OMe)_3$
(C) $HF < HI$ (D) $NH_3 > SbH_3$
17. Which of the following options is/are **INCORRECT** :-
- (A) $O < N < F$ (IE_2 order)
(B) $O^{-2} < O^{-1} < O^{+} < O$ (E.A order)
(C) $Li_2O < BeO < NO_2 < N_2O_5$ (acidic nature of oxide)
(D) $Ni < Cu < Zn$ (atomic size)

SECTION-IV

- This section contains 7 questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. Find the number of chemical species(s) which are planar and nonpolar in the following :-
 Al_2Cl_6 , NO_2 , O_2F_2 , XeF_5^+ , HCN , XeF_5^- , $B_3N_3H_6$, B_2H_6 (2)
2. Which oxy acid have M = O and M-O-H both type of bonds (M = Central atom)
Sulphurous acid ; Arsenous acid ; Hypochlorous acid ; Perchloric acid ;
Peroxynitric acid ; Pyrosulphuric acid (3)
3. How many of the following have at least two same atoms connected in a continuous manner, in their respective structure?
 $H_4P_2O_7$, $H_2S_2O_5$, $H_2S_2O_6$, Cl_2O_7 , $H_2S_2O_8$, P_4O_{10} , S_3O_9 , H_3PO_5 (4)
4. Which of the following pairs have dipole-dipole interaction
 $CHCl_3$, CCl_4 , HBr , Br_2 , ICl (3)
5. Find the number of molecules, which have bond angle more than 100°
 NH_3 , PH_3 , SH_2 , AsH_3 , H_2Se , SiH_4 (2)
6. Find the number of chemical species which on complete hydrolysis form one of the product as dibasic oxyacid :-
 PCl_3 , BCl_3 , SOE_4 , TeF_6 , P_4O_6 (4)
7. What is minimum value of 'n' possible for an electron having : $m = -3$ and $s = -\frac{1}{2}$? (4)

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IITIANS CAREER

INORGANIC CHEMISTRY

TIME : 30 MIN.

SECTION-I (I)

- This section contains **10** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases

Which of the following option is **CORRECT** for indicate property ? (M = underline atom)

- (A) $\underline{N}(\text{SiH}_3)_3 < \underline{P}(\text{SiH}_3)_3$: (Si- $\hat{M}$ -Si B.A.)
- (B) $\underline{N}(\text{SiH}_3)_3 < \underline{P}(\text{SiH}_3)_3$: (Back bonding)
- (C) $\underline{N}(\text{SiH}_3)_3 < \underline{P}(\text{SiH}_3)_3$: (Si-M Length)
- (D) $\underline{N}(\text{SiH}_3)_3 < \underline{P}(\text{SiH}_3)_3$: (π -character in Si-M bond)

Silicate in which discrete tetrahedral unit is present are also called **neso silicate**.

Identify example of neso silicate :-

- (A) $\text{BaTi}(\text{Si}_3\text{O}_9)$ (B) $\text{Sc}_2\text{Si}_2\text{O}_7$
- (C) $\text{Ca}_2\text{Mg}_5(\text{Si}_4\text{O}_{11})_2(\text{OH})_2$ (D) Be_2SiO_4

If size of two metal ions of boron family are :

$$M_a^+ = 1.20 \text{ \AA}$$

$$M_b^+ = 1.40 \text{ \AA}$$

Then select the **CORRECT** for size

- (A) Size of $M_a^{3+} > M_b^{3+}$ ✗ (B) Size of $M_a^{3+} > M_b^+$ ✗
- (C) Size of $M_b^+ > M_b^{3+}$ ✗ (D) Size of $M_b^+ > M_b^{3+} > M_a^{3+}$

Select molecule in which presence of lone pair(s) on central atom does not affect ideal bond angle as per hybridisation :

- (A) XeO_3 (B) BrF_5 (C) XeF_4 (D) ClF_3

Select **INCORRECT** order of BF_3 and BBr_3 :

- (A) Bond angle : $\text{BBr}_3 > \text{BF}_3$ ✓ (B) Lewis acidic strength : $\text{BBr}_3 > \text{BF}_3$
- (C) π -character in B-X bond : $\text{BF}_3 > \text{BBr}_3$ ✓ (D) Boiling point : $\text{BF}_3 < \text{BBr}_3$

Select which of the following is monobasic acid but does not act as proton donar :

- (A) CsOH (B) HOCl (C) H_3PO_2 (D) H_3BO_3

Number of oxygen atoms bonded with one P atom, one Si atom & one C atom respectively in P_2O_6 , SiO_2 & CO_2 : $O=C=O$ (2)

- (A) 3, 4, 2 (B) 6, 2, 2 (C) 3, 2, 2 (D) 3, 2, 4

Which of the following molecules do not exist ?

- (A) BiI_3 (B) SH_4 (C) BrF_6^- (D) Both (A) and (B)

Which of the following is a good oxidising agent ?

- (A) Pb^{+4} (B) Pb^{+2} (C) Ga^{+3} (D) Sn^{+4}

10. If internuclear axis is z-axis, then, in which of the following cases σ and π bonds are not possible ?

- (A) $d_{xy} + d_{xz}$ (B) $p_y + p_y$ (C) $d_{z^2} + d_{z^2}$ (D) $p_z + p_z$

SECTION - I (II)

- This section contains 3 questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :

+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 In all other cases

11. Lewis theory fails to explain which of the following structure(s) ?

- (A) SF_6 (B) XeF_2 (C) XeF_4 (D) $BeCl_2$

Hydrogen bonding is responsible for

- (A) Layer structure of solid boric acid
(B) Less volatile nature of p-nitrophenol as compared to its ortho isomer
(C) More solubility of CH_3CH_2OH in water as compared to CH_3-O-CH_3

OH

(D) Higher boiling point of fluorobenzene as compared to benzene

12. Which of the following produce 'ic' acid of central atom on hydrolysis :-

- (A) PCl_3 (B) P_4O_{10} (C) SF_6 (D) BrF_5

SECTION-I(IV) Matching List Type

- This section contains **3** questions.
- Each question has **matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

14. List - I (Molecule)

- (P) $O(SiH_3)_2$
(Q) B_2H_6
(R) $H_3BO_{3(s)}$
(S) SO_2

List - II (Characteristic)

- (1) Banana bonding
(2) Hydrogen bonding
(3) $p\pi-d\pi$ back bonding
(4) Both $p\pi-p\pi$ and $p\pi-d\pi$ bonding

Code :

	(P)	(Q)	(R)	(S)
(A)	4	1	2	3
(C)	2	1	3	4

	(P)	(Q)	(R)	(S)
(B)	3	1	2	4
(D)	1	2	3	4

15. List - I

(Molecule)

- (P) Marshall's acid ($H_2S_2O_8$)
(Q) Triphosphoric acid ($H_3P_3O_{10}$)
(R) Copper sulphate ($CuSO_4 \cdot 5H_2O$)
(S) Cynuric trichloride ($C_3N_3Cl_3$)

List - II

(Properties)

- (1) Ionic bond present
(2) Cyclic with six membered ring
(3) Contains peroxy linkage
(4) Contains "X-O-X" linkage

Code :

	(P)	(Q)	(R)	(S)
(A)	2	1	4	3
(C)	4	1	2	3

	(P)	(Q)	(R)	(S)
(B)	3	4	1	2
(D)	4	3	2	1

16. List - I

(Species)

- (P) SF_4
(Q) Crystalline HF
(R) SF_6
(S) O-nitrophenol

List - II

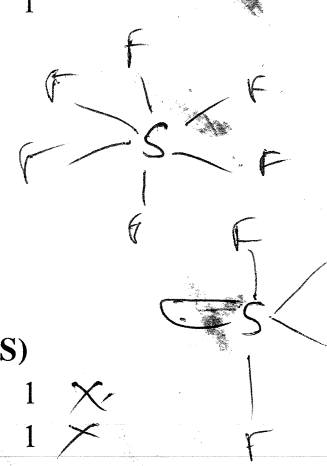
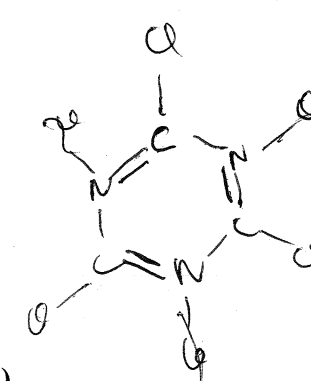
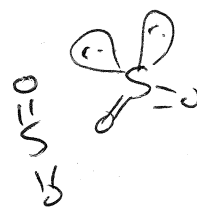
(Properties)

- (A) Inter-molecular H-bonds
(B) sp^3d hybridization
(C) Intra-molecular H-bond
(D) Octahedral

Code :

	(P)	(Q)	(R)	(S)
(A)	2	1	4	3
(C)	4	1	2	3

	(P)	(Q)	(R)	(S)
(B)	2	3	4	1
(D)	4	3	2	1



SECTION-II : Matrix-Match Type

This Section contains **1 question**. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the following column :

Column-I

(Molecules)

(A) ICl (P) (R) (T)
(B) C/Cl (Q) (R) (T)
(C) NH_3 (P) (R)
(D) N_2 (Q) (R) (S) (T)

Column-II

(Characteristic)

- (P) Polar bond(s)
(Q) Non polar bond(s)
(R) Sigma bond(s)
(S) Pie bond(s)
(T) Number of lone pair in molecule > one

SECTION-IV

- This section contains **3 questions**.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :

+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened

0 If none of the bubbles is darkened

-1 In all other cases

Find the number of molecules in which axial orbital length of central atom is higher than equatorial orbital length of central atom



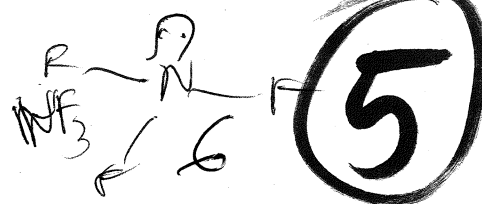
(5)

Find the maximum number of molecules which give(s) product acid on hydrolysis from their central atom having similar basicity.



(4)

Find the number of chemical species which does NOT exist :-



(5)

SECTION-I : (ii) ONE OR MORE OPTIONS CORRECT TYPE

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the bubble(s) corresponding to the correct option(s) is (are) darkened.

Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, Provided NO incorrect option is darkened.

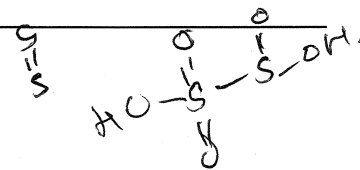
Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -2 In all other cases.

- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in -2 marks, as a wrong option is also darkened

1. Identify the **CORRECT** statement :-

- (A) Peroxy linkage is present in marshall's acid
- (B) Pyrophosphoric acid on hydrolysis gives phosphoric acid
- (C) Basicity of phosphorous acid is two
- (D) Debye force is responsible for the formation of clathrate compound of Xe in ice



2. Which of the following order is/are **CORRECT** :-

- (A) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$ (Lattice energy)
- (B) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$ (Melting point)
- (C) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$ (Polarizing power of cation)
- (D) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$ (% ionic character)

3. Which of the following statement is/are **CORRECT** :-

- (A) In noble gases helium has lowest boiling point
- (B) In $\text{NaHCO}_3(\text{s})$ hydrogen bonding takes place
- (C) Boiling point of SiCl_4 is greater than the boiling point of CCl_4
- (D) Boiling point of H_2O is greater than the boiling point of HF .

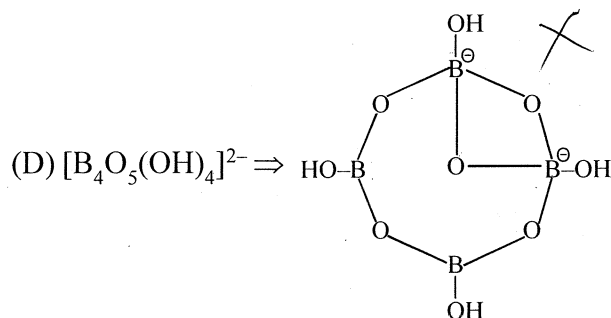
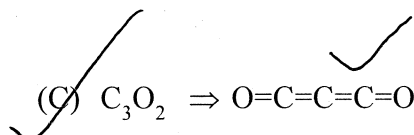
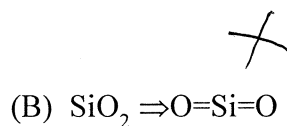
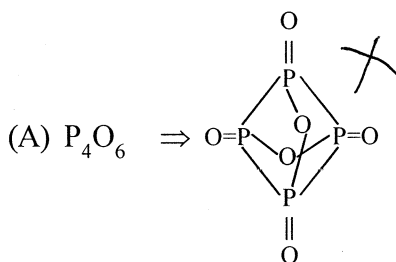
4. Select the **CORRECT** order against the mentioned property :-

- (A) $\text{NaNO}_3 < \text{KNO}_3 < \text{RbNO}_3$ (Thermal stability)
- (B) $\text{NaF} > \text{KF} > \text{RbF}$ (Covalent nature)
- (C) $\text{Si-O} < \text{P-O} < \text{S-O} < \text{Cl-O}$ (π -bond strength)
- (D) $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$ (Bond length)

5. Which base among the following reacts with diborane to cleave it to give $[\text{BH}_4]^-$

- (A) T.H.F.
- (B) Me_3N
- (C) $\text{C}_5\text{H}_5\text{N}$
- (D) MeNH_2

6. Select the **CORRECT** structure :



SECTION-I (I)

- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
 - +3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
 - 0 If none of the bubbles is darkened
 - 1 In all other cases

7. Which of the following molecule is most covalent :-

- (A) $ZnCl_2$ (B) $CdCl_2$ (C) $HgCl_2$ (D) $CuCl$

8. Which of the following option is **CORRECT** for indicated property ?

- (A) $O(CH_3)_2 > O(SiH_3)_3$: X-O-X (Bond angle)

- (B) $N(SiH_3)_3 < P(SiH_3)_3$: (Extent of Back bonding)

- (C) $SOF_2 > SOCl_2$: S-O (Bond length)

- (D) Diamond < Graphite :

(Thermodynamic stability)

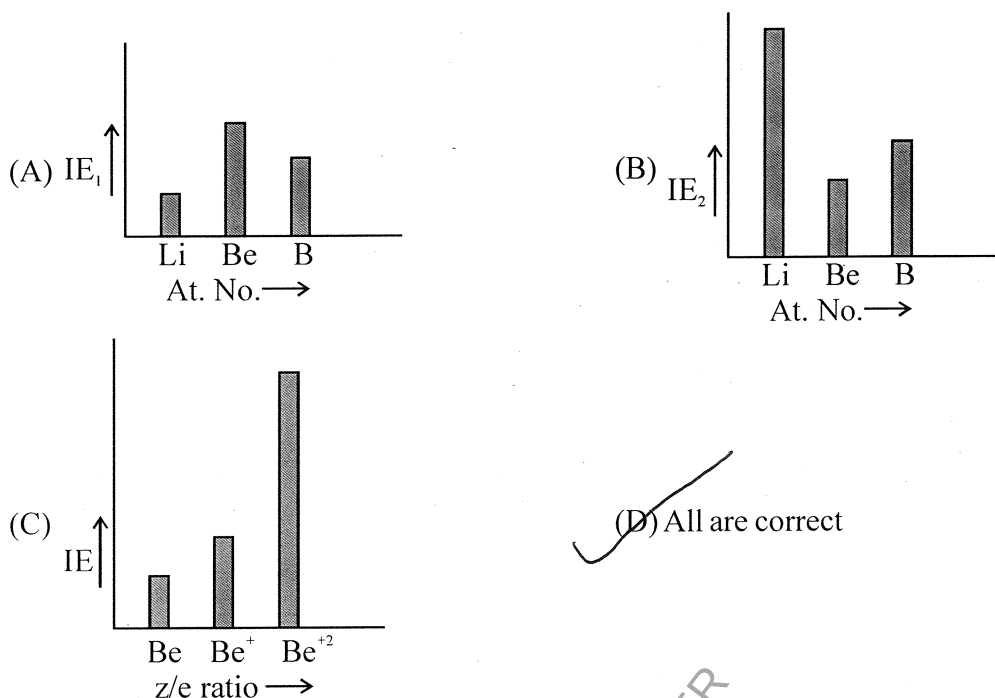
9. Which of the following have different number of $p_\pi - d_\pi$ bonds present as compared to other molecule ?

- (A) SO_2 (B) $SOCl_2$ (C) $HClO_3$ (D) H_2SO_3

10. How many elements of the 4th period have only one electron in orbital with $n=4, l=0, m=0$ in their ground state electronic configuration -

- (A) One (B) Two (C) Three (D) Twelve

11. Select **CORRECT** graphical representation for relative comparison of Ionisation Energy (I.E.)

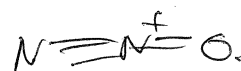


12. Select the molecule in which d-orbital(s) is involved in hybridisation as well as in π -bond formation with oxygen

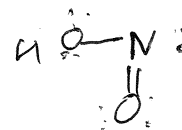
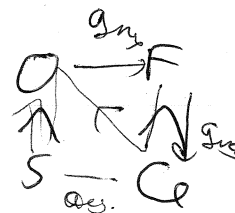
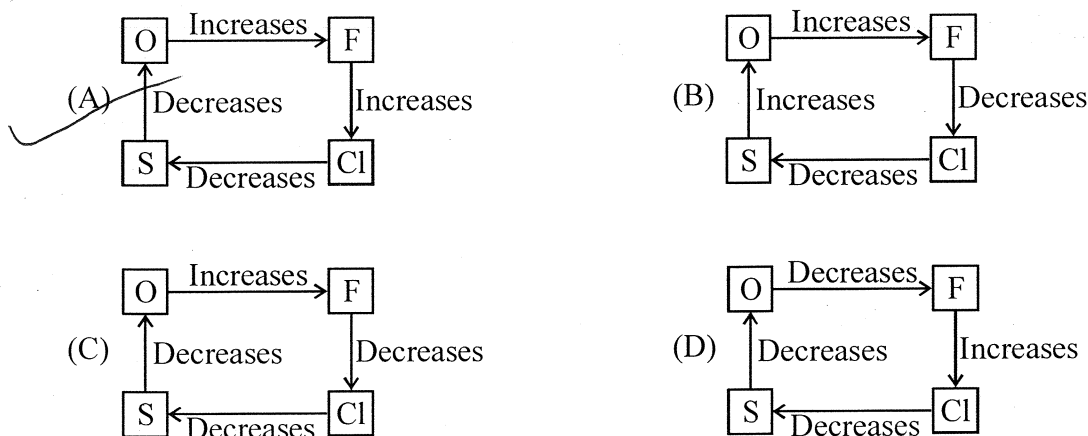
(A) SO_2F_2 (B) SOF_4 (C) SO_2 (D) SOF_2

13. Which of the following is written with its **CORRECT** characteristics

(A) C_3O_2 : Non planar, Non-polar (B) $H_2C=C=O$: Planar, Polar
(C) $H_2C=C=CH_2$: Planar, Non-polar (D) NNO : Planar, Non-Polar



14. Select **CORRECT** variation for the electron affinity of O, F, S, Cl



15. Select the pair which has difference of only one neutron

(A) ^{14}N and ^{15}O (B) ^{12}C and ^{16}O (C) ^{12}C and ^{14}C (D) ^{14}N and ^{15}N

16. Find total number of lone pairs in C_2N_2 and HNO_2 respectively :

(A) 4, 5 (B) 2, 4 (C) 5, 4 (D) 2, 5

17. Select **INCORRECT** order of stability :

- (A) $Pb^{+2} > Pb^{+4}$ (B) $Sn^{+4} > Pb^{+4}$ (C) $Pb^{+2} > Sn^{+2}$ ~~(D) $Sn^{+2} > Sn^{+4}$~~

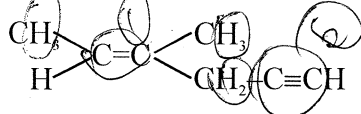
18. Consider the following two statements -

(I) Electronic configuration of C = $1s^2 2s^1 2p^3$ is more stable as compared to C = $1s^2 2s^2 2p^2$ due to half filled stability of p-subshell *incor.*

(II) In COF_2 , orbital of carbon which forms π -bond has 33.33% s-character

- (A) I- correct - II -correct (B) I- incorrect - II -correct
(C) I- correct - II -incorrect ~~(D) I- incorrect - II -incorrect~~

19. Find number of sp^3 -s, σ -bonds and $2p_\pi-2p_\pi$ bonds respectively in



- (A) 5, 3 (B) 6, 2 ~~(C) 8, 3~~ (D) 8, 2

20. The screening constant for the last electron in Cr^{+5} :

- (A) 24 (B) 18 (C) 23 ~~(D) 22~~

SECTION-II : Matrix-Match Type

This Section contains **1 question**. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (P, Q, R, S & T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. Match the following column :

Column-I

(Atom)

- (A) Se *S T Q*
(B) Br *S R*
(C) Xe *Q T*
(D) B *Q R*

Column-II

(Characteristic)

- (P) have vacant p-orbital in outer shell
(Q) Can have covalency = two
(R) Can have covalency = three
(S) 4th shell is outer most shell
(T) Produce MF_4 type fluorides molecule (M - given atom)

SECTION-IV

- This section contains **1** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. Identify the number of chemical species having ionic character less than LiI :-

LiCl, NaCl, NaBr, KI, CsI, NaI

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SECTION-I (I)

- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
+3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. Which of the following have $3c-2e^-$ and $2c-2e^-$ bond as well as non planar geometry ?

- (A) $(BeH_2)_n$ (B) $(BeCl_2)_n$ (C) Be_2H_4 (D) $Al_2(CH_3)_6$

2. Which of the following order is **CORRECT** with respect to the given property ?

- (A) $PbCl_4 > PbCl_2$: Stability
(B) $Pb^{+2} > Pb^{+4}$: (Oxidising strength)
(C) $Sn^{+4} > Sn^{+2}$: (Reducing strength)
(D) $AgCl > NaCl > KCl$: Covalent character

3. Which of the following molecule is non planar with respect to underlined atom ?

- (A) $(SiH_3)_3\text{P}$ (B) $GeH_3-\underline{N}CO$ (C) $(GeH_3)_3\underline{N}$ (D) $[B_3O_6]^{3-}$

4. Which oxy acid have only two $p_\pi - d_\pi$ bonds in structure :-

- (A) Pyrosulphuric acid (B) Orthophosphoric acid
(C) Hypophosphorous acid (D) Peroxysulphuric acid

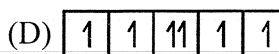
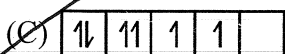
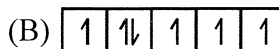
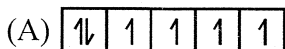
5. Consider the following elements with their electronegativity value.

Element	X	Y	Z	W
Electronegativity	3.77	1.12	2.25	3.10

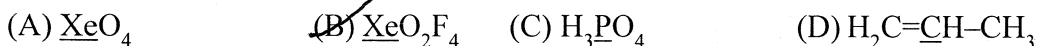
(Pauling scale)

INCORRECT statement is :

- (A) $X-OH$ is more acidic than WOH
(B) YOH is more basic than ZOH
(C) XY molecule is predominantly ionic
(D) ' $W-OH$ ' bond is more weaker than ' $Y-OH$ ' bond in polar solvent
6. Which of the following electronic configuration violate Hund's rule for a d^6 system of an isolated gaseous species ?



7. Which of the following molecule has minimum % s-character in its hybrid orbital for the underlined atom :



SECTION-I : (ii) ONE OR MORE OPTIONS CORRECT TYPE

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the bubble(s) corresponding to the correct option(s) is (are) darkened.

Partial Marks : +1 For darkening a bubble corresponding to **each correct option**, Provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

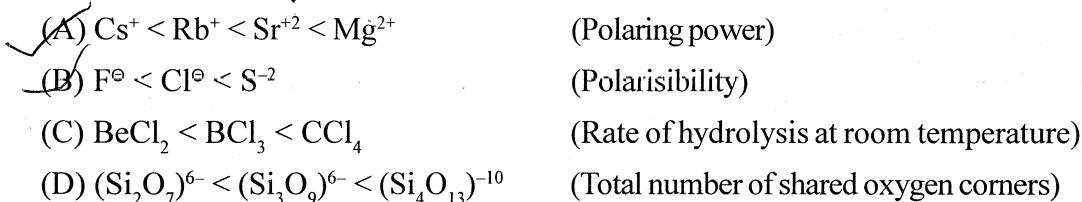
Negative Marks : -2 In all other cases.

- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in -2 marks, as a wrong option is also darkened

8. In which of the following silicate each tetrahedral unit have two unshared oxygen ?



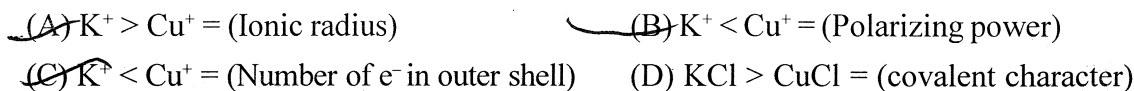
9. Which of the following order is/are **CORRECT** for the property indicated against it :-



10. Which of the following does have pyramidal shape.



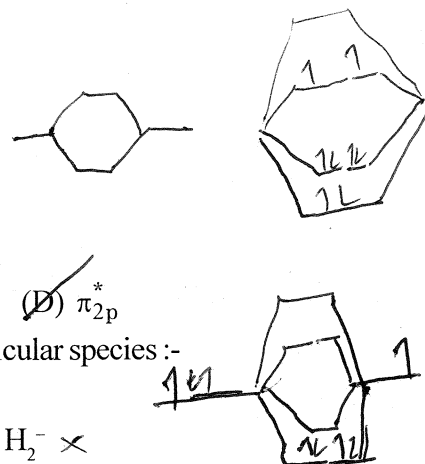
11. Identify the **CORRECT** order :-



12. The **CORRECT** order of match for the energy distance function for following interaction.



13. In which of the following case the magnetic behaviour get changed :-
~~(A)~~ $B_2 \rightarrow B_2^{+2}$ ~~(B)~~ $C_2 \rightarrow C_2^{+}$ (C) $C_2 \rightarrow C_2^{2-}$ ~~(D)~~ $B_2 \rightarrow B_2^{2-}$
14. Which of the following is/are **CORRECT** order :
~~(A)~~ $LiF > NaF > KF > RbF > CsF$ (Thermal stability)
 (B) $MgF_2 > MgO > Mg_3N_2$: (Lattice energy) ✓
 (C) $Be(NO_3)_2 > Mg(NO_3)_2 > Ca(NO_3)_2$: (Thermal stability)
 (D) $BaO > CaO > ZnO > GeO$: (Basic strength)
15. In which of the following option(s) heat of vaporization is increases on moving left to right.
 (A) HF, HCl, HBr ✓ (B) CH_4, NH_3, H_2O
 (C) H_2O, H_2S, H_2Se (D) CH_4, SiH_4, GeH_4
16. δ -bonding molecular orbital can be formed by :-
 (A) $d_{xy} + d_{xy}$ when overlap on x-axis
~~(B)~~ $d_{xy} + d_{xy}$ when overlap on z-axis
~~(C)~~ $d_{x^2-y^2} + d_{x^2-y^2}$ when overlap on z-axis
 (D) $d_{xz} + d_{xz}$ when overlap on z-axis
17. Which molecular orbital of O_2 is/are doubly degenerated :-
 (A) σ_{2p} (B) σ_{2s}^* ~~(C)~~ π_{2p} ~~(D)~~ π_{2p}^*
18. Which electron is at highest energy among all the electrons of that particular species :-
~~(A)~~ Unpaired electron of atomic boron (B) Unpaired electron of B_2
 (C) π_{2py} electron of molecular oxygen (O_2) (D) Pair of electron in H_2^- ✗



SECTION-I(iii) : (PARAGRAPH TYPE)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there will be **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme :
 +3 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
 0 In all other cases

Paragraph for Q.No. 19 to 20

When a substance undergoes nucleophilic substitution reaction and the nucleophile is the solvent itself, then the reaction is known as solvolysis. In solvolysis reaction, if the solvent used is water then the reaction is called as hydrolysis. Hydrolysis can occur through several mechanism.

19. Which of the following compound produce only one type of acid as a product on hydrolysis ?
 (A) N_2O_4 ~~(B)~~ PCl_5 (C) NCl_3 ~~(D)~~ $AsCl_3$
20. Which of the following statement is **INCORRECT** ?
 (A) During the hydrolysis of $SiCl_4$, T.S. is formed which have T.B.P. geometry at central atom
 (B) Equimolar amount of PCl_5 and water show partial hydrolysis and produce two mole hydra acid
~~(C)~~ Hydrolysis product of BCl_3 have basicity is 3
 (D) SiF_4 undergoes partial hydrolysis but BCl_3 undergoes complete hydrolysis

Paragraph for Q.No. 21 to 22

As its name implies, **Molecular Orbital Theory** allocates electrons to **molecular orbitals** similar in many respects to atomic orbitals, but each containing two or more nuclei. To derive the ground state of the molecule, the electrons are placed in the available molecular orbitals, beginning with the orbital of lowest energy; each molecular orbital can accommodate a maximum of two electrons, provided their spins are opposed (Pauli's principle extended to molecules); and by filling as few orbitals as possible, and those of the lowest energy possible, we arrive at the electronic ground state.

21. N_2 molecule is diamagnetic and have 946 KJ/mole bond energy and bond length 1.10 Å, when N_2 molecule is ionized to unipositive ion :-

- (A) Bond energy increases
(B) Bond length decreases
(C) $\Delta H_{(IE)}$ for given change is higher than $\Delta H_{(IE)}$ of N
(D) Product remain diamagnetic

22. If each atomic orbital and molecular orbital can hold maximum 3 electrons. Then select **CORRECT** statement(s) :-

- (I) Electronic configuration of O = $1s^3, 2s^3, 2p^2$ ✓
(II) O_2 molecule have total 10 bonding electrons ✓
(III) O_2 molecule have total 8 antibonding electrons ✓

(A) I, II

(B) II, III

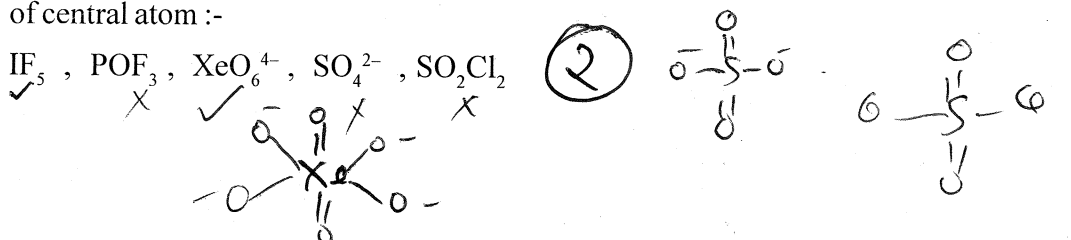
(C) I, III

(D) I, II, III

SECTION-IV (INTEGER TYPE)

- This section contains 2 questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme :
+4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened
0 If none of the bubbles is darkened
-1 In all other cases

1. Find the total number of ions which have higher hydration energy as compared to Sr^{2+} :-
 $Rb^+, Ba^{+2}, Ca^{+2}, Al^{+3}, Cs^+, Be^{+2}, Sc^{+3}$ (5)
2. Find the total number of allotropes contain discrete molecules held together by London forces :-
Diamond, Graphite, Fullerene, White phosphorus, Rhombic sulphur, Black phosphorus (3)
3. Find the total number of molecules/ions in which at least one 'd' orbital is commonly used in hybridisation of central atom :-



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SECTION-I : (i) Only One option correct Type

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. 3(-1)

1. Decomposition of which salt required highest thermal energy.
 (A) Carbonate of barium ☒ (B) Carbonate of zinc
 (C) Carbonate of calcium ☒ (D) Carbonate of mercury ☒
2. If $M^+X^- > M^+Y^- > M^+Z^-$ (Covalent character)
 $M^+X^- < N^+X^-$ (Ionic character)
 Select cation with highest polarising power and anion with highest polarisability respectively :-
 (A) M^+ , X^- ☒ (B) N^+ , X^- (C) M^+ , Z^- (D) N^+ , Z^-
3. Saturated solution of which fluoride salt have maximum concentration of fluoride ion.
 (A) LiF (B) NaF (C) RbF ☒ (D) CsF
4. Which of the following transformation, the bond order has increased and magnetic behaviour gets changed :-
 (A) $NO^+ \rightarrow NO$ (B) $N_2 \rightarrow N_2^-$ (C) $O_2^{+2} \rightarrow O_2$ ☒ (D) $CO \rightarrow CO^+$
5. Which of the following molecular species has unpaired electron(s)
 (A) O^{2-} ☒ (B) N_2^+ (C) C_2 ☒ (D) B_2^{2-} ☒
6. In which of the following process the value of magnetic moment does not change
 (A) $O_2^+ \rightarrow O_2^{+2}$ (B) $N_2 \rightarrow N_2^-$ (C) $CO \rightarrow CO^+$ ☒ (D) $B_2^+ \rightarrow B_2^-$
7. In which type of interaction doubling the distance between the molecules reduce the strength of their interaction energy by the factor of 8 :-
 (A) Ionic interaction ☒ (B) Ion-dipole interaction ☒
 (C) dipole-dipole interaction (D) London interaction ☒

SECTION-I : (ii) One or more options correct Type

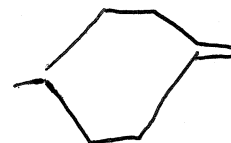
This section contains **13 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

8. Which of the following statement(s) is/are **INCORRECT** :-
 (A) B_2 has one sigma bond ☒ (B) $2s-2p$ orbital mixing is not possible in N_2 ☒
 (C) C_2 having 1σ and 1π -bond bond ☒ (D) Bond order of F_2 same as O_2^{2-} ☒
9. Which of the following statement(s) is/are **CORRECT**.
 (A) $O_2 \rightarrow O_2^+$ is oxidation process and $O_2 \rightarrow O_2^-$ is reduction. ☒
 (B) F_2 is pale yellow due to homo-lumo transition between π^* and π M.O electrons ☒
 (C) O_2, F_2 are formed from by no s-p mixing ☒
 (D) C_2, B_2 are formed by s-p mixing ☒

10. Consider given electronic configuration (ground state) in which s-p mixing is not operative
 $X_2 = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \dots = \dots, \dots = \dots, \sigma^* 2p_x^0$
 If bond order of X_2 is = one.

Select **CORRECT** for missing molecular orbitals.

- (A) All can be vacant
 (B) All have same energy
 (C) All can have some number of electron
 (D) Two orbital of lower energy are full filled and other two are vacant



11. Select correct for He_2^+

- (A) It is called helium molecule cation
 (B) Overall one electron is stabilize
 (C) It has half filled σ_{1s}^*
 (D) It has half filled σ_{1s}

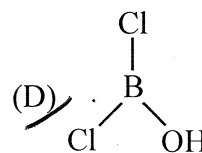
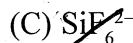
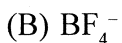
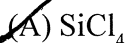
12. Which of the following have same electronic distribution in orbitals (Consider MOT for molecule or molecular ion)

- (A) O and F
 (B) O^{2-} and F^-
 (C) O_2 and F_2
 (D) O_2^{2-} and F_2

13. Which of following molecule(s) is/are having π -bond only :-

- (A) N_2^{+2}
 (B) B_2
 (C) C_2^{-2}
 (D) O_2

14. Select which can produce extra co-ordinate bond with H_2O to form transition state when react with water :-



15. Which have σ -type coordinate bond :-

- (A) BF_3
 (B) BF_4^-
 (C) NH_2^-
 (D) O_3

16. Which of the following species are diamagnetic in nature.

- (A) O_2
 (B) N_2
 (C) B_2
 (D) NO

17. Which of the following is **CORRECT** order of bond energy of σ -bond :

- (A) $N-N > P-P$
 (B) $F-F < Cl-Cl$
 (C) $C-C > Si-Si$
 (D) $O-O < S-S$

18. Which of the following species does not exist at room temperature ?

- (A) P_2
 (B) S_2
 (C) OF_4
 (D) NCl_5

19. Which of the following species is/are involve d_{z^2} orbital in bonding.

- (A) SF_4
 (B) SOF_4
 (C) $XeF_5^\ominus$
 (D) XeO_3F_2

20. Which of the following molecule(s) is/are planar :-

- (A) $ICl_4^\ominus$
 (B) XeF_2
 (C) $XeF_5^\ominus$
 (D) $I_3^\oplus$

SECTION-II : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

(B) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

(C) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

(D) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).

- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 In all other cases

1. List (I)

- (A) $\text{Ba}(\text{OH})_2$ **PQ T**
(B) $\text{Ca}(\text{OH})_2$ **T**
(C) $\text{Mg}(\text{OH})_2$ **TS**
(D) $\text{Be}(\text{OH})_2$ **R**

List (II)

- (P) most soluble
(Q) most basic
(R) produce salt with NaOH_{Aq}
(S) Milk of magnesia
(T) Produce salt with $\text{H}_2\text{SO}_4_{\text{Aq}}$

2. List (I)

(A) $s + s$

$\xrightarrow{\text{BMO}}$

P T
(B) $p_y + p_y$

$\xrightarrow[\text{internuclear axis (x)}]{\text{ABMO}}$

(Q) Ungerade symmetry

QS (C) $d_{xy} + d_{xy}$

$\xrightarrow[\text{internuclear axis (z)}]{\text{BMO}}$

(R) Odd number of nodal plane

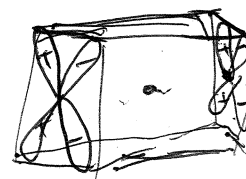
(D) $d_{xy} + d_{xy}$

$\xrightarrow[\text{internuclear axis (x)}]{\text{BMO}}$

(S) 4 lobes interaction by each orbital

QT

(T) σ - molecular orbital



SECTION-IV : (Integer Value Correct Type)

This section contains **6 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive)

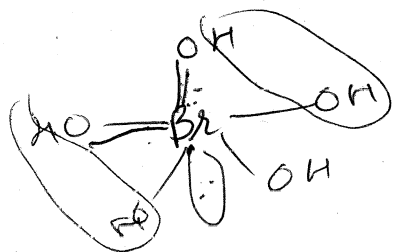
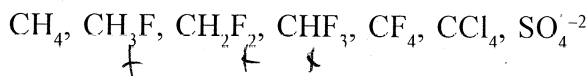
4(-1)

- ~~3.~~ Species having same bond order and magnetic behaviour :-
 NO^+ , CN^- , CO , N_2 , F_2 , O_2^{+2} (4)
- ~~2.~~ Find number of $p_\pi - d_\pi$ bonds in bromine containing product of the hydrolysis of BrF_5 (2)
- ~~1.~~ Find total number of unshared 'O' corner(s) in pyrosilicate ion. (2)
4. Select number of properties which can be correctly represented by following order. (7)

$$\text{LiBr} > \text{CsBr}$$

Covalent character ; lattice energy ; hydration energy ; Thermal stability ; Solubility in water ;
Inter-nuclear distance

- Find maximum number of electrons in degenerated molecular orbitals of O_2 .
6. Find number of molecules is not perfect tetrahedral geometry.



$$qE = mg \quad \text{and} \quad qE = 6\pi\eta Rv$$

$$6\pi\eta Rv = mg \Rightarrow v = \frac{mg}{6\pi\eta R}$$

$$v = \frac{4}{3} \pi R^2 \rho g \Rightarrow \frac{4}{3} \pi R^2 \rho g = \frac{mg}{6\pi\eta R}$$

$$\Rightarrow R = \sqrt{\frac{9mg}{8\pi\eta\rho g}}$$

$$Q_E \neq 6\pi \eta V_T \sqrt{\frac{q n V_T}{2 T g}}$$

$$9.8 \times 10^5 - 18 \times 10^{-6} \times 2 \times 10^{-3}$$

$$q = 8 \times 10^{-19}$$

$$\frac{4.3 \times 10^{-5} \times 2 \times 10^{-3}}{\sqrt{49.98 \times 2 \times 900}} \times 9$$